



# ***BLENDER SECRETS***

Create a City with fog  
using Blender 4 and fSpy

by Jan van den Hemel

# Why this e-book?

Hi! Allow me to introduce myself. **My name is Jan** and you may know me from my “[Daily Blender Tips](#)”. The reason I started making these tips is because even as a professional Blender user, I still sometimes need to look stuff up (“what was that shortcut again?”) and could not find a good source of really short tutorials.

For things you only do once in a while (like baking normal maps), even the professionals will sometimes go to Youtube to patiently endure a 40-minute tutorial where the actual explanation comes at the 37-minute mark. Wouldn't it be great if these topics were addressed in under 1 minute?

I couldn't find any channel with short, one-minute video tutorials for Blender, so I decided to make them myself. Of course, not everything can be shown in 1 minute, but a surprising amount of things can! The Blender Secrets e-book collects these Blender tips in written form, so you can learn them at your own pace.

I hope it will serve you well and that you will enjoy reading it.

**This 30-page e-Book will hopefully give you a good idea of what to expect in the full, almost 2000-page (and always growing) e-Book.**



Best wishes,  
Jan

## Create a mysterious city with Blender and fSpy



I had the opportunity to take some photographs from the Tokyo Metropolitan Building, which are useful for this tutorial. You can download these photos [here](#). Let's turn one of these photos into a dark, mysterious and dystopian 3D landscape and render a short animation.



To do this we'll use **fSpy**, which is both a free stand-alone app and a free Blender add-on. You'll need the app to create the fSpy files, and the add-on to import them in Blender.

I've noticed from the comments on [older tutorials](#) about this topic, that a lot of people download the **wrong version** of fSpy and therefore think it no longer works. So please make sure you use the links for the correct versions, found on the next page.

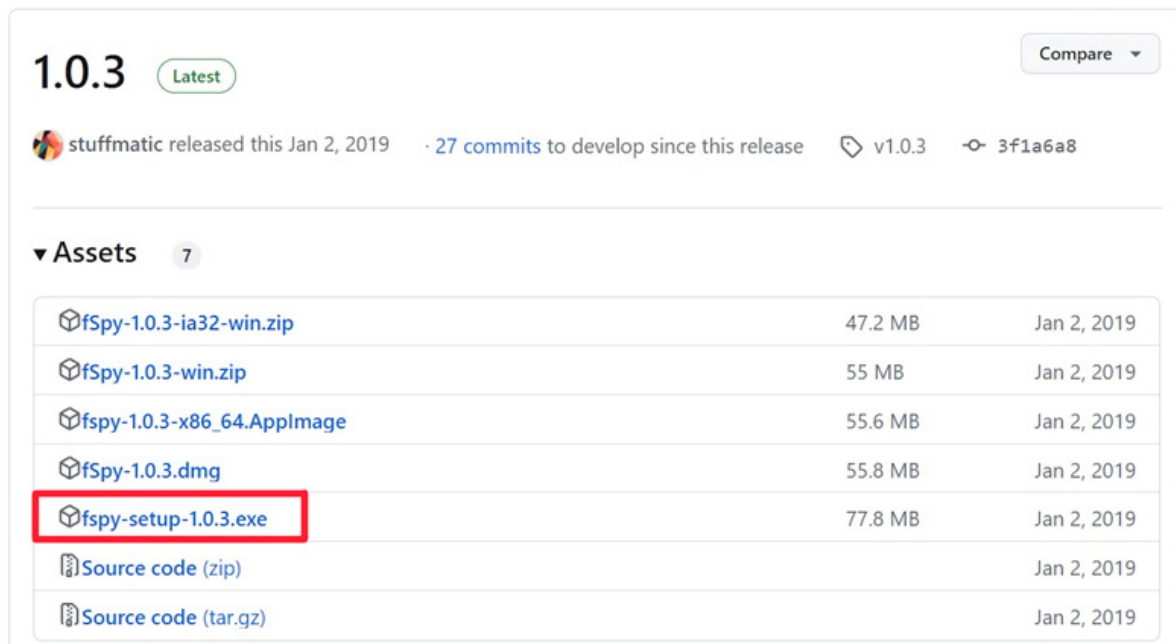
## Download fSpy stand-alone app:

<https://github.com/stuffmatic/fSpy/releases/tag/v1.0.3>

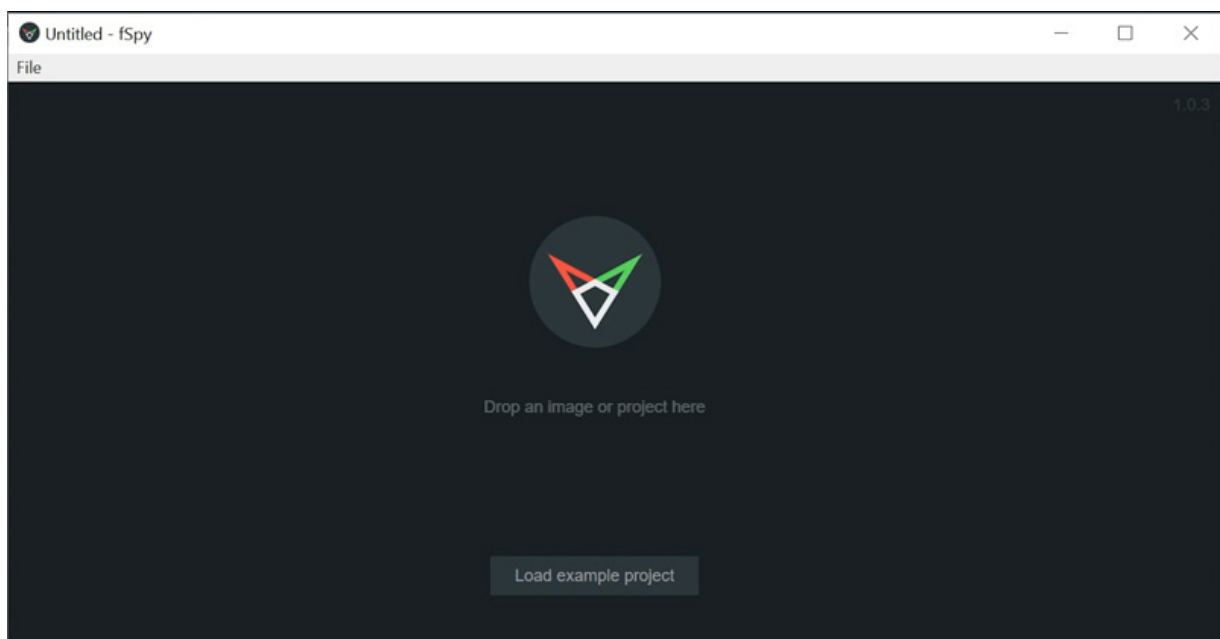
## Download fSpy add-on:

<https://github.com/stuffmatic/fSpy-Blender/releases/tag/v1.0.3>

To get started, download the fspy setup file, **fspy-setup.1.0.3.exe**, from Github. Then click on the downloaded file to install it.



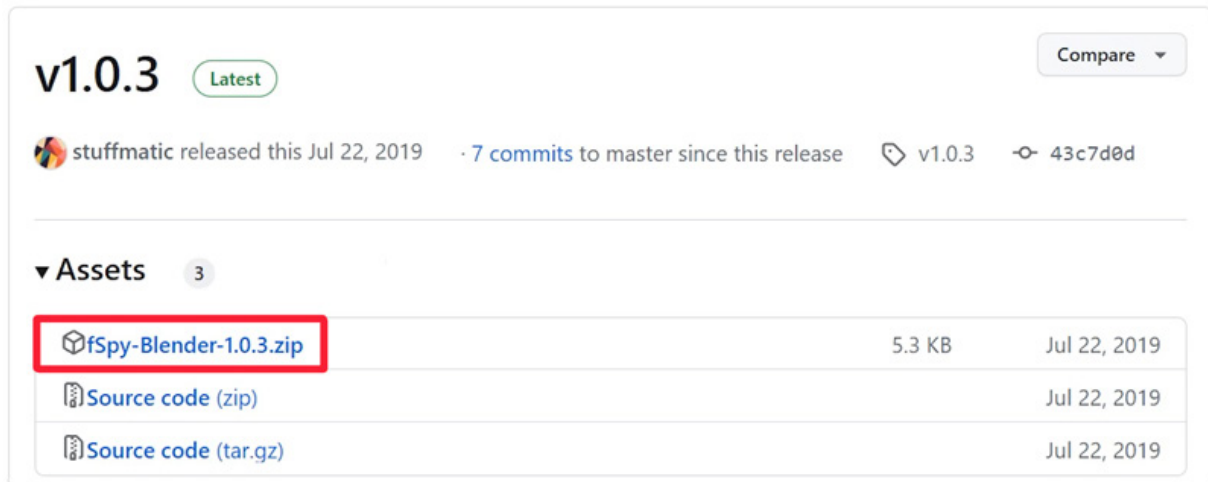
Below you see the interface of the **stand-alone fSpy app**. In this window, you can drag and drop a photo to get started (more in that later). It's a pretty simple app to use.



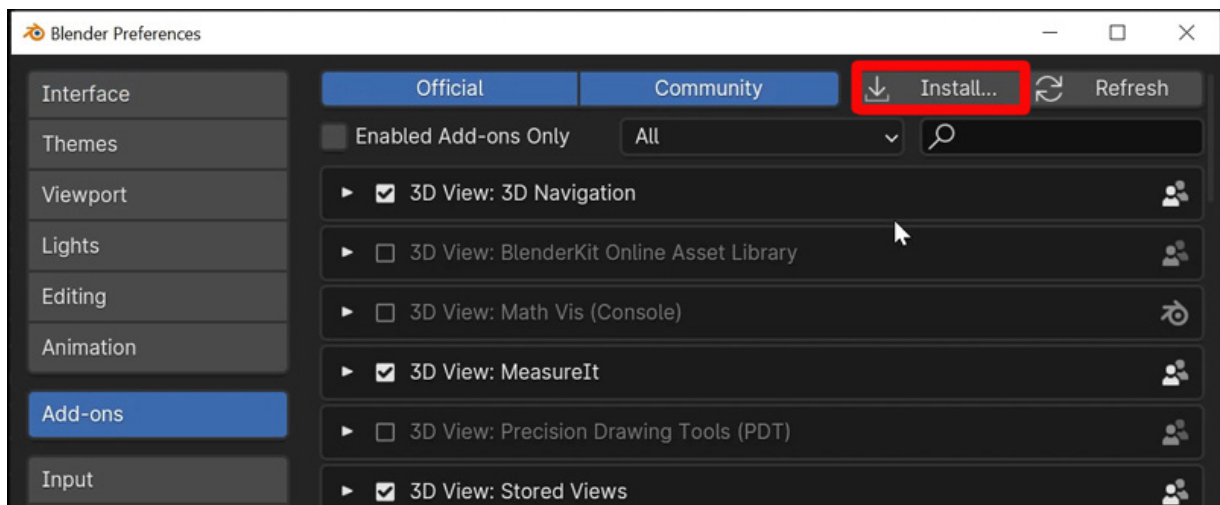
You'll also need to download the [Blender fSpy add-on, version 1.0.3](#).

Without this add-on you won't be able to import the files made in the fSpy app into Blender.

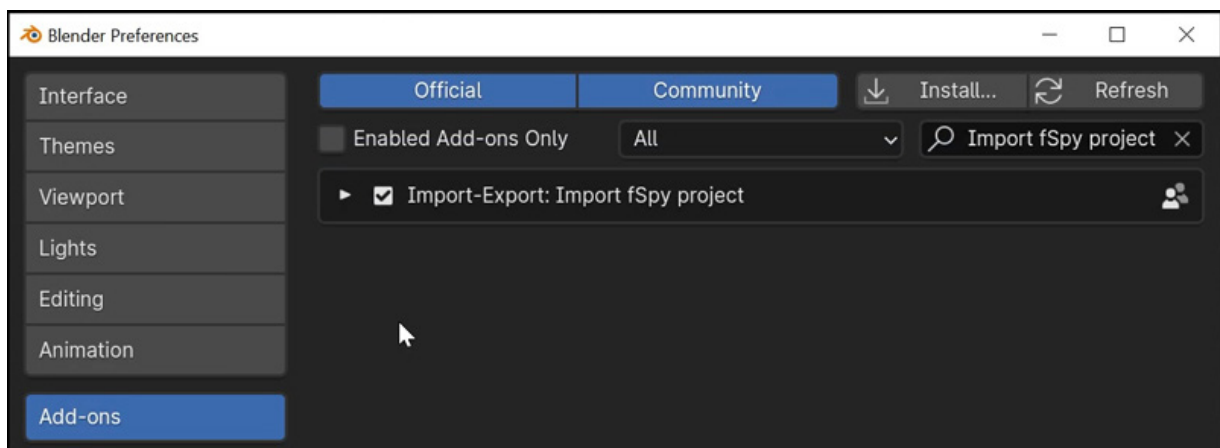
Download the ZIP-file, but don't unzip it! This goes for most blender add-ons that you download from the web.



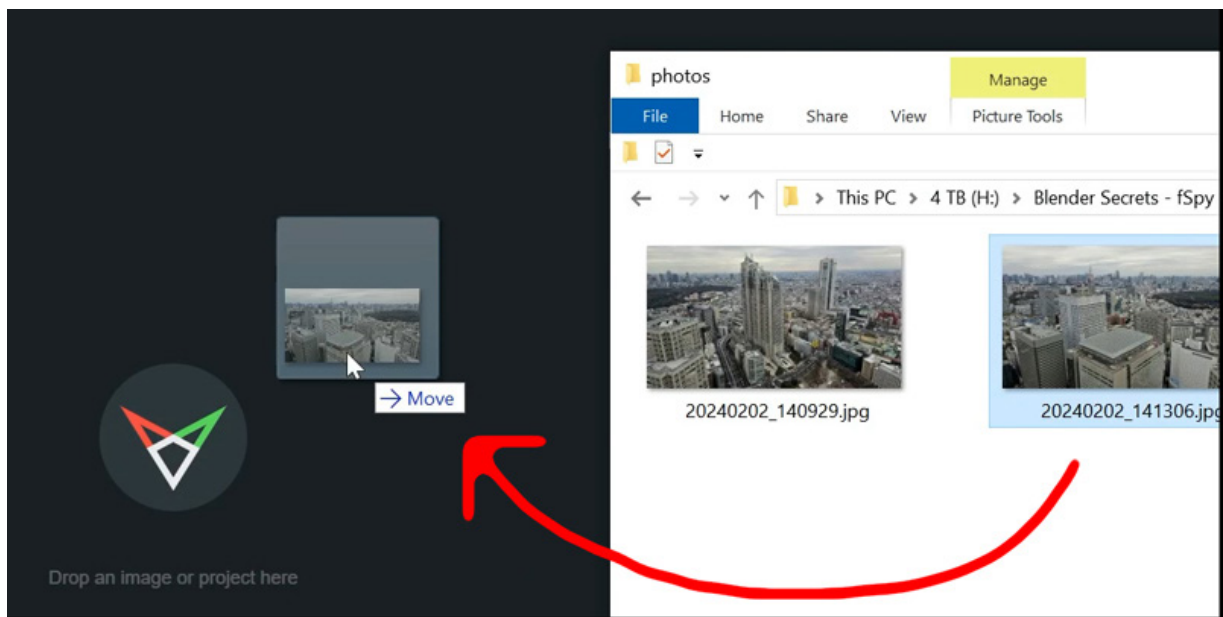
In Blender, go to **Preferences** and install the add-on. Click on **Install** and select the downloaded ZIP file.



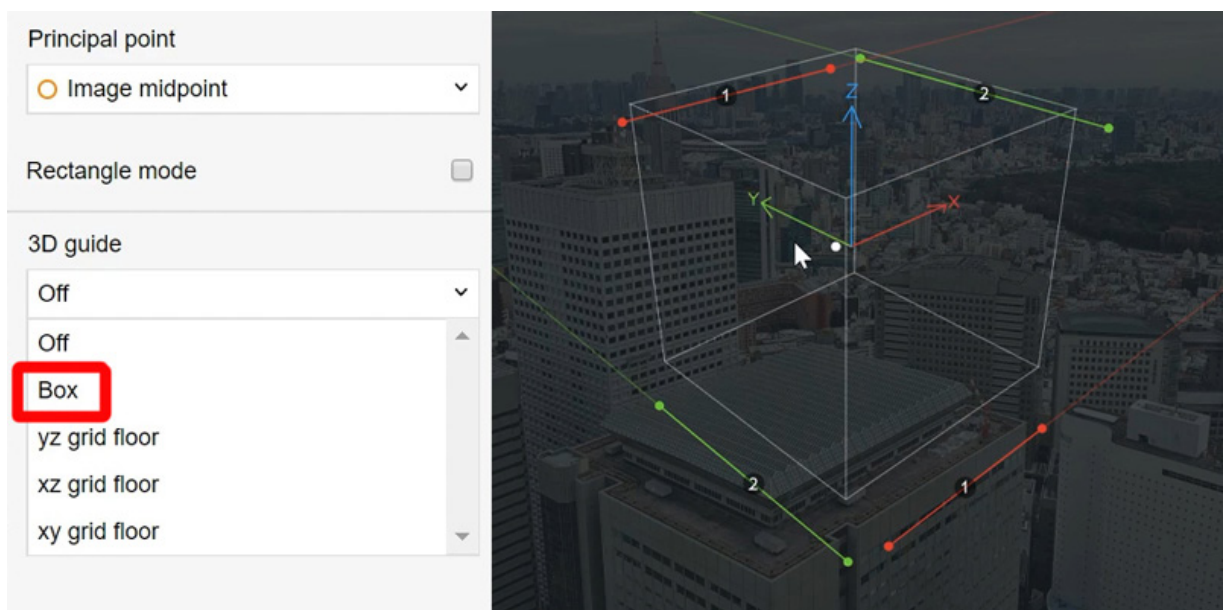
And finally enable it. You may have to search for it in the add-ons list.



Drag and drop a photograph into the stand-alone fSpy app.



We can enable a **3D guide** in the shape of a **box**. You can move this cube around to check if the perspective lines up with the photo. At first it won't line up, so we'll have to adjust the vanishing points.

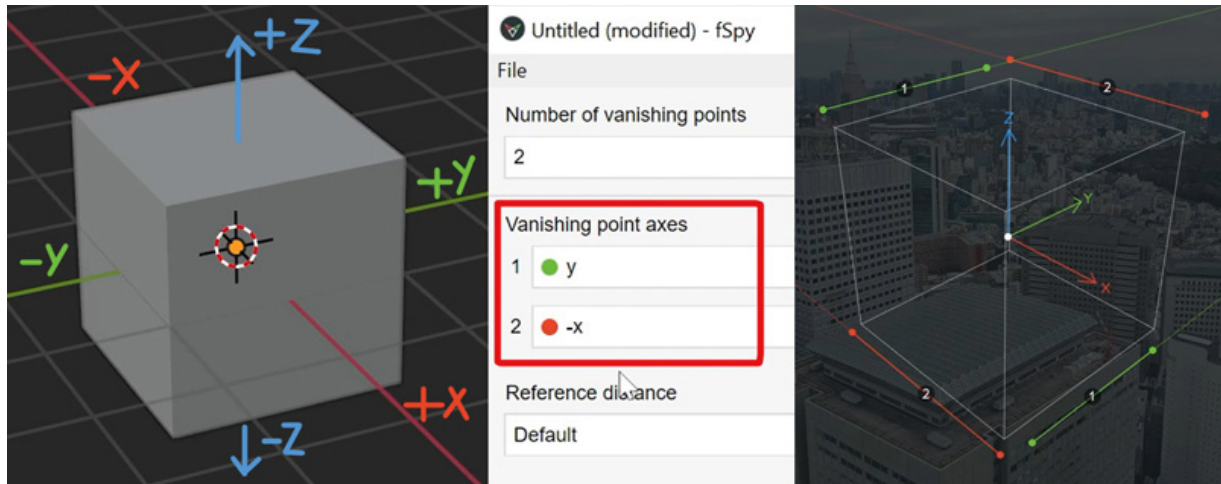


I would like to have the **X, Y and Z-axes** in fSpy point in the same direction as in Blender.

That way we won't end up with some weirdly oriented upside down scene in Blender. For reference, you can see the axis in the Navigation Gizmo in Blender.

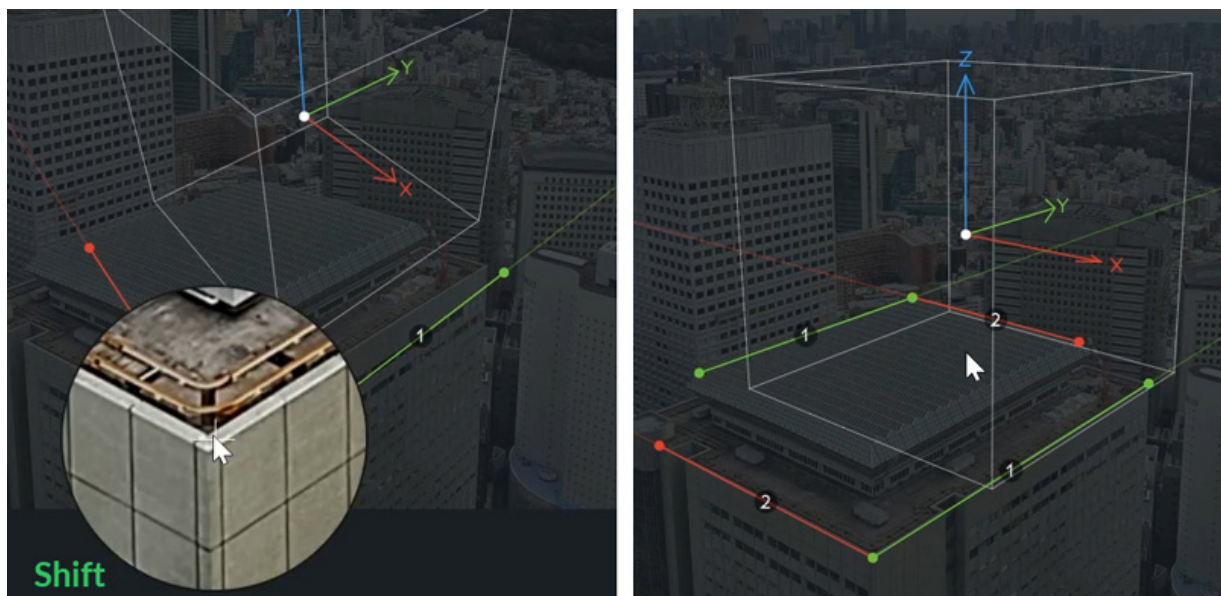
To make the axes in fSpy correspond to those in Blender, we will need to change the **vanishing point axes**. To y and -x in this specific case (this depends on your photo, you'll need to experiment).

As you can see, now they match those in Blender.



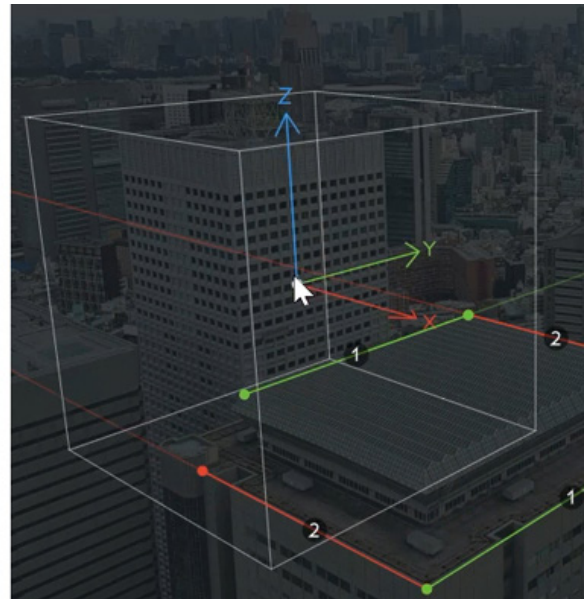
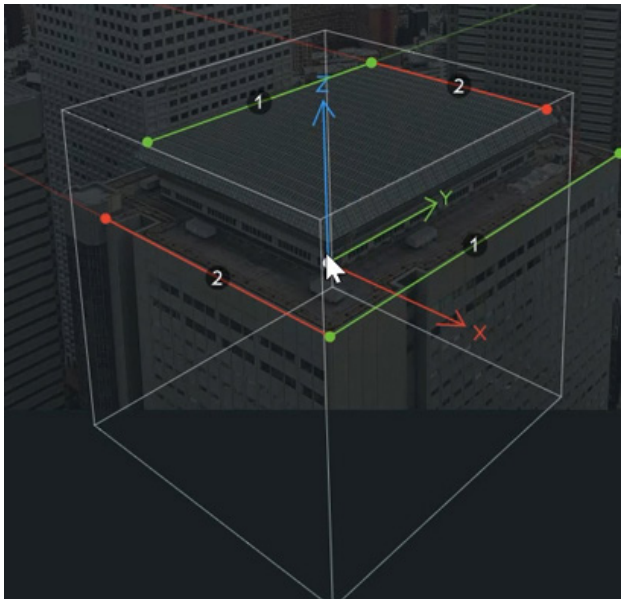
Now we can start aligning the vanishing points, by **clicking on and then dragging their end points**.

By **holding Shift**, you can zoom in to more precisely place them. It looks a bit like you're looking through a sniper rifle scope...



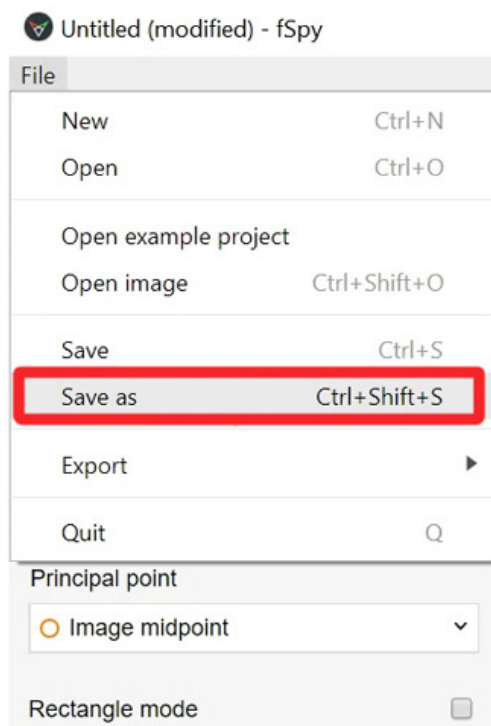
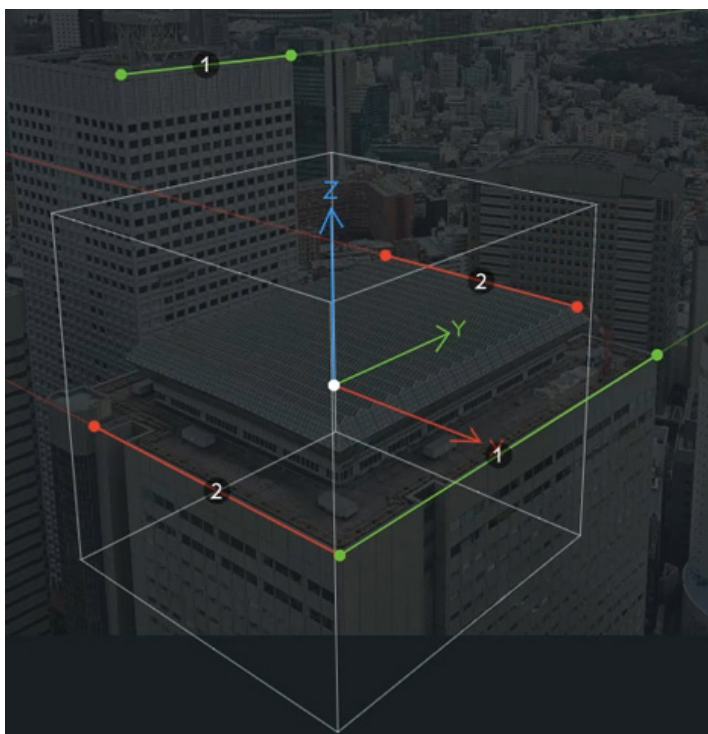
In this case, I'm using the building in front as reference as it has some clear perspective lines.

We can test if the perspective of the **3D guide cube** matches the buildings. It seems to be almost perfect. But not quite, as the Z-axis still seems to be a little different than the vertical edge of the building.



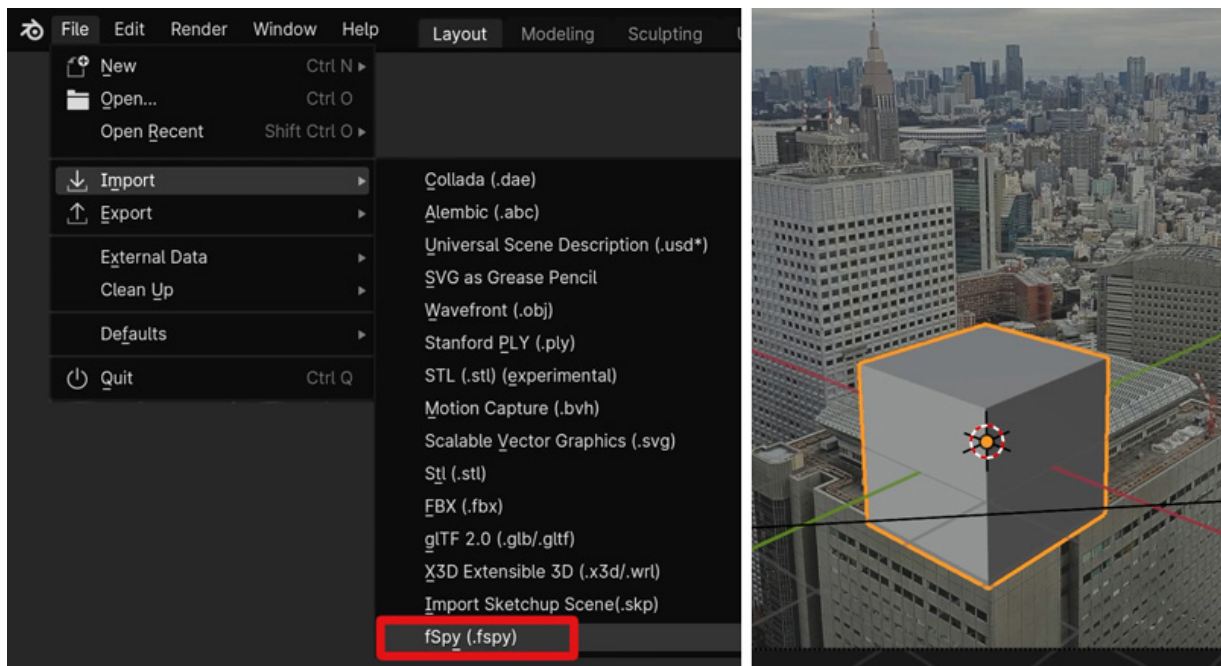
I think we can improve it by using a building further away as reference for this vanishing point. Now it seems to align even better. The next step is to **save this file**.

This file can then be imported in Blender.



Because we installed the fSpy add-on, we now have an fSpy import option. I'll import the file I just saved from the stand-alone app by going to **File > Import > fSpy** and choosing the correct file.

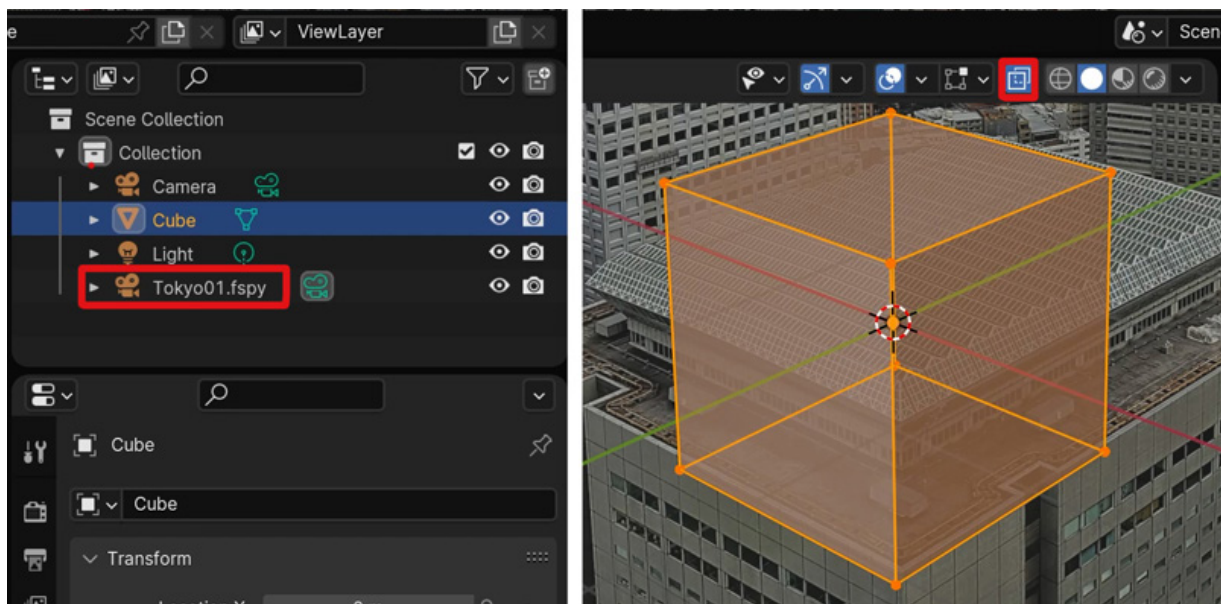
This file contains the photograph as well as a camera. And as you can see, from the Camera view, the Cube aligns to the perspective of the photo. Press 0 on the Numpad to enter Camera view, in case you exit it by accident.



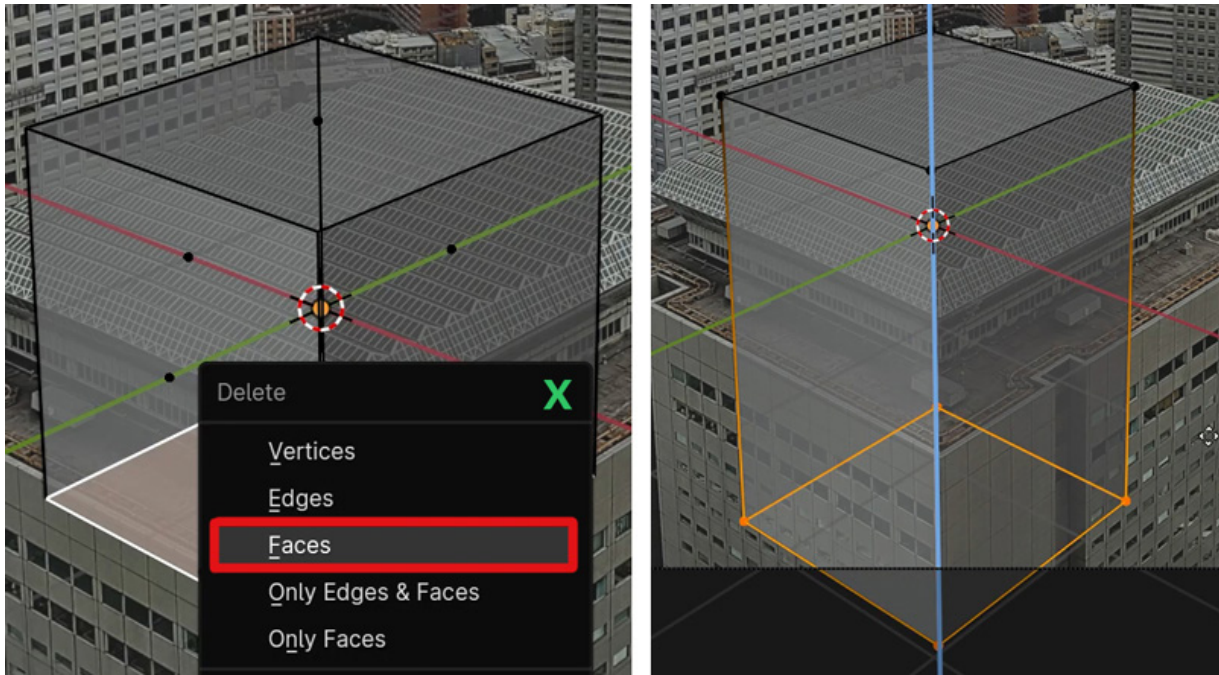
It's a good idea to save this .blend-file at this point.

We still have the original **Camera** from the default Blender scene, which we don't need. So select that one and just delete it.

As you can see the imported Camera has the name of the fSpy project, "Tokyo01.fspy". I'll turn on **X-Ray mode** so I can see the buildings in the photograph through the mesh. This will make modeling easier.

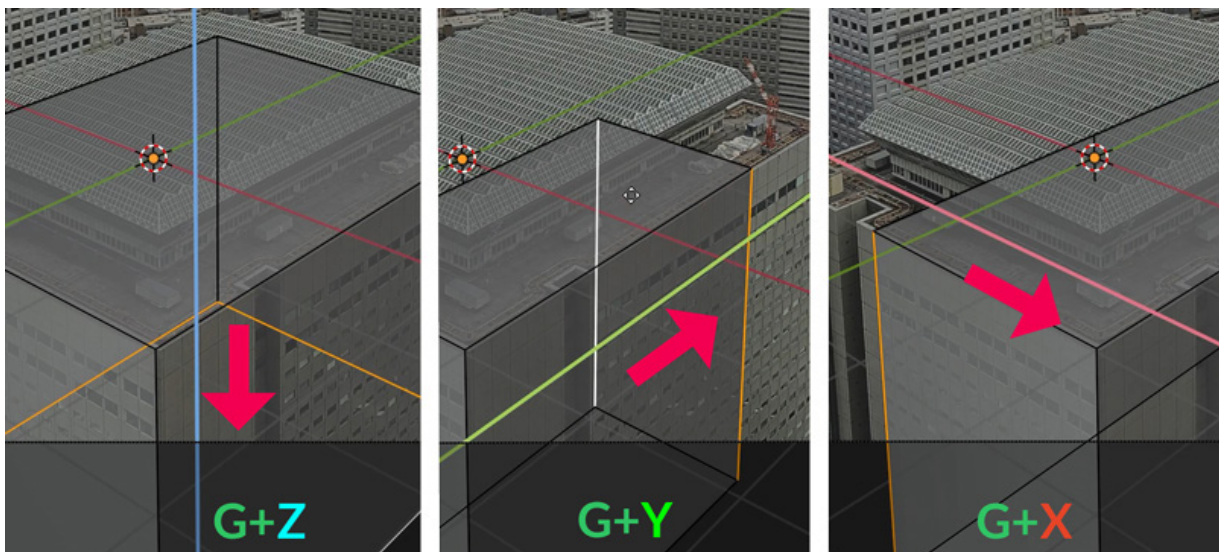


And in **Edit Mode** I'll delete the bottom Face as it won't be visible. Now it's just a matter of moving Vertices, to match the buildings in the photo.



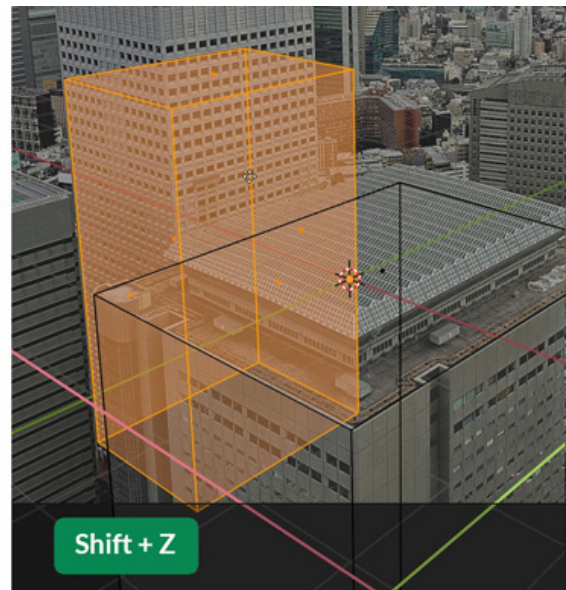
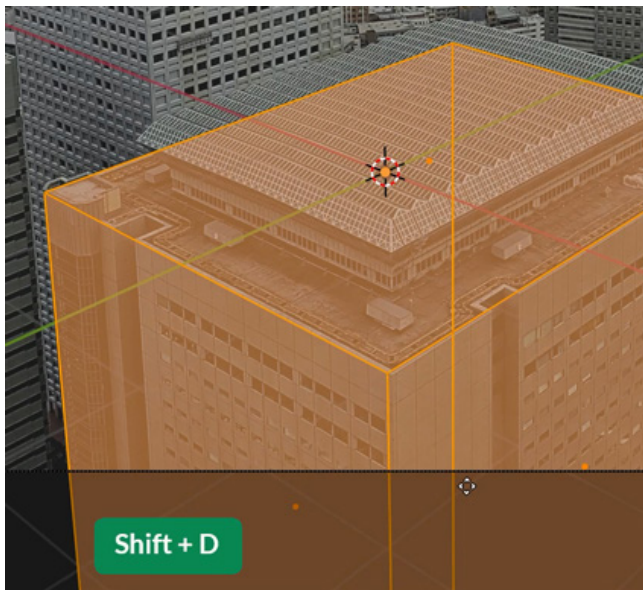
By holding **Alt** and clicking, we can select all these bottom **Edges** and move them down.

To move things along just one axis, press for example G and Z to move it along the Z-axis. Or G and Y to move along the Y-axis... or G and X to move along the X-axis.

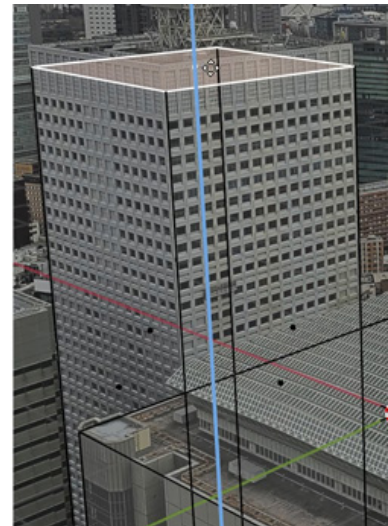
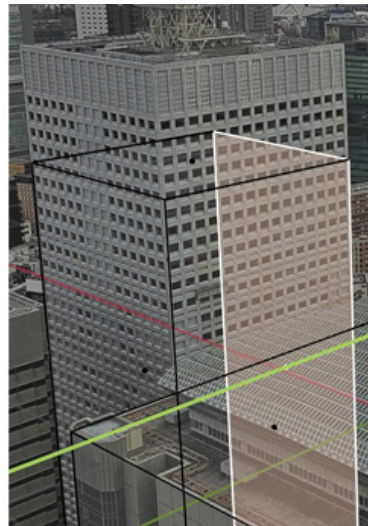
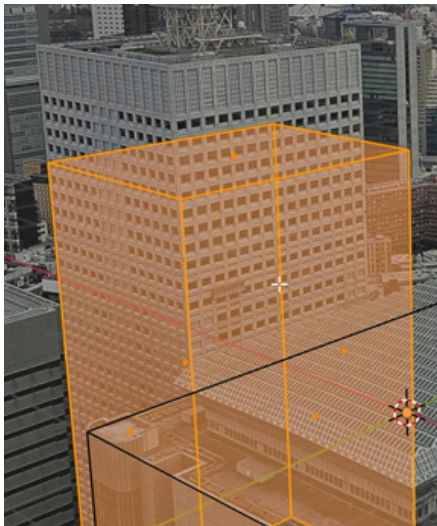


This is the very basic shape of this building, from here on it's just a matter of adding more detail.

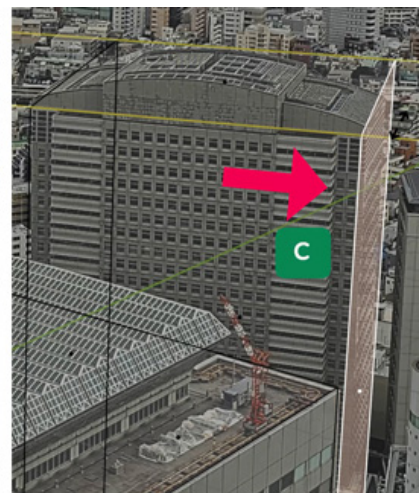
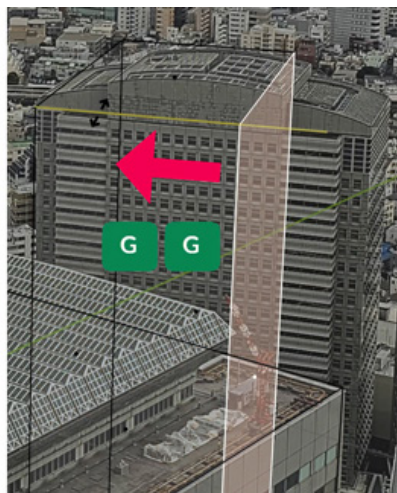
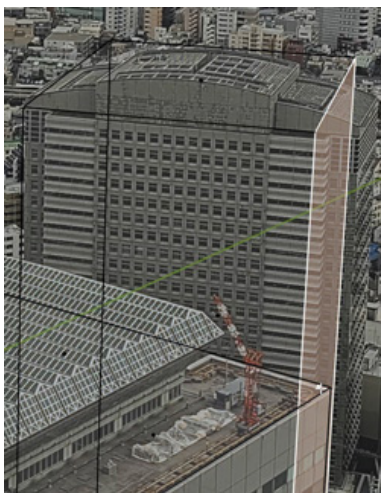
But first let's select all and duplicate with **Shift+D**, for the next building. By pressing G to move and then **Shift+Z**, we exclude the Z-axis so we can move it easily without accidentally moving up or down.



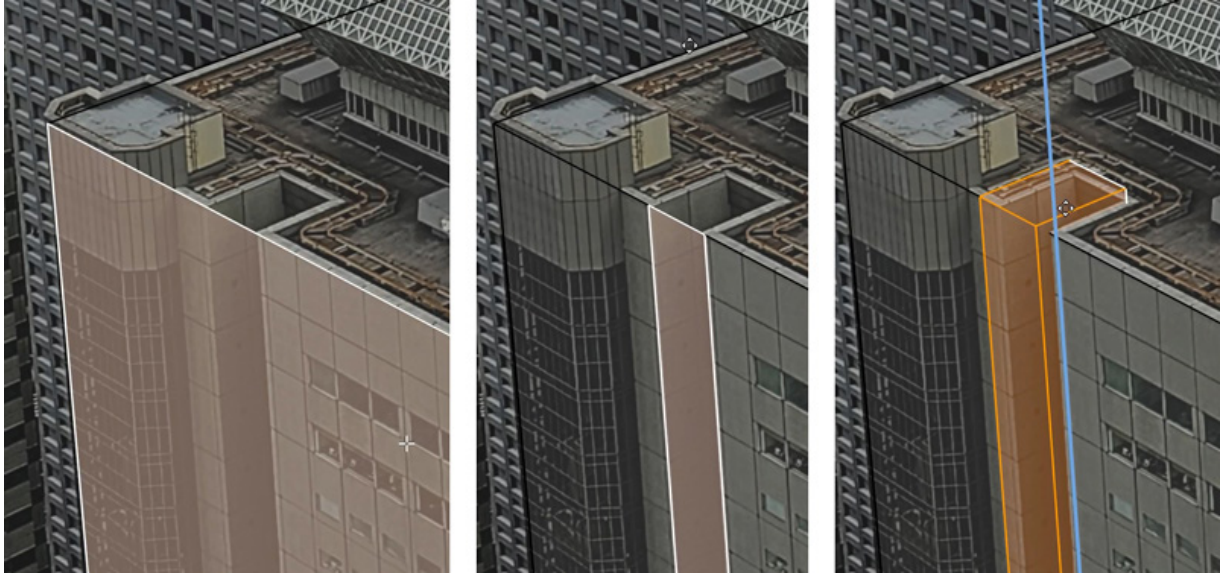
And then it's just a matter of adjusting the shape by moving Faces.



If you **press G twice**, select an Edge to slide along - it will turn yellow - **and then C**, you can slide 'outwards' along that Edge. This can be convenient if you've rotated something in Edit Mode, and need to move along its local orientation.

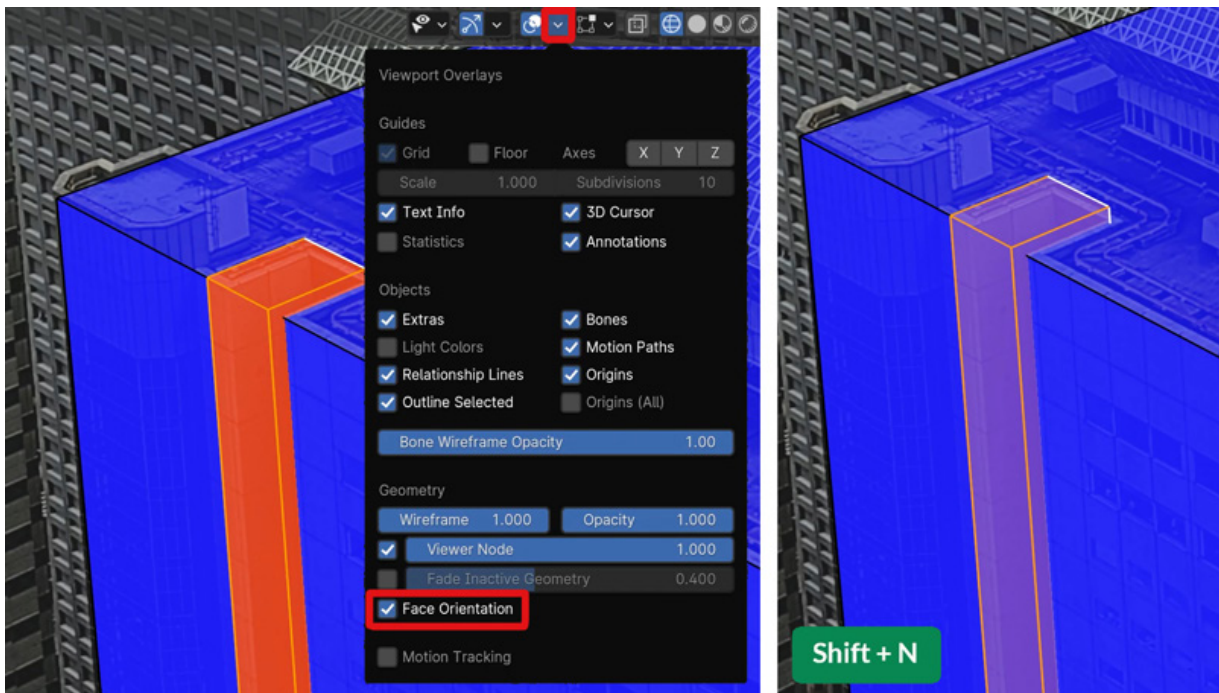


To create these gaps in the side of the building, I'll select and duplicate the **Face** at the side where the gap should be. Then scale it along the X-axis and move it to make it match the width of the gap. Then extrude along the Y-axis with E and Y. Move it slightly so there are no Faces overlapping in the exact same location (that would cause issues later).

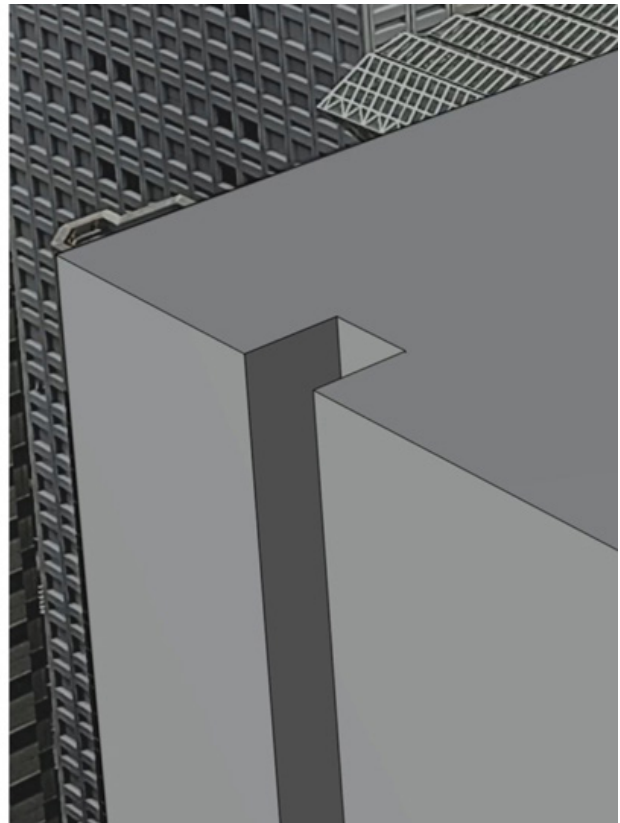
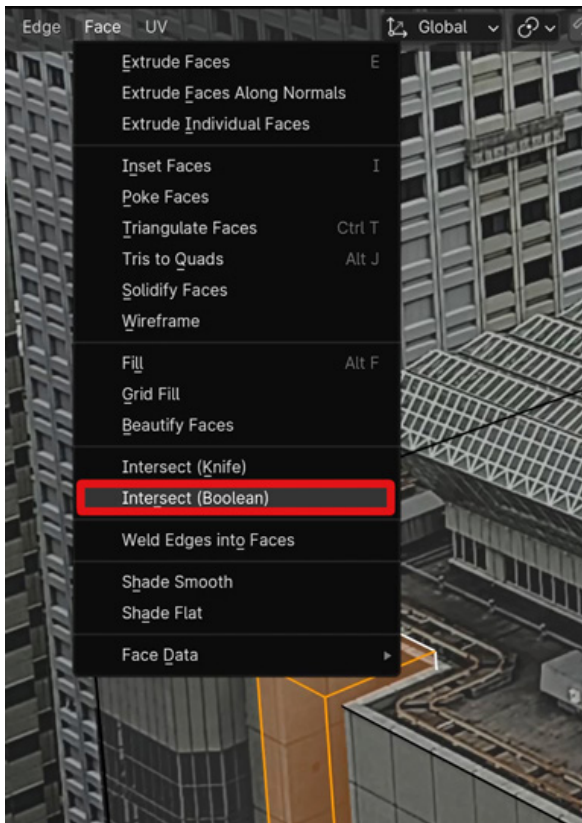


A quick check of the **Face Orientation**, shows that the Faces of this geometry are Flipped (shown in red).

Pressing **Shift+N** recalculates the normals so they're no longer flipped. I'm planning to do a boolean cut, and flipped Faces can cause trouble.

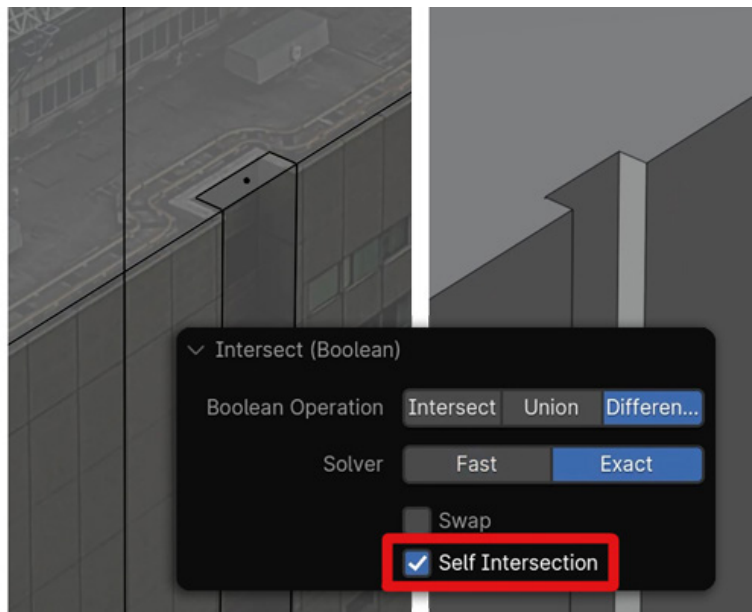
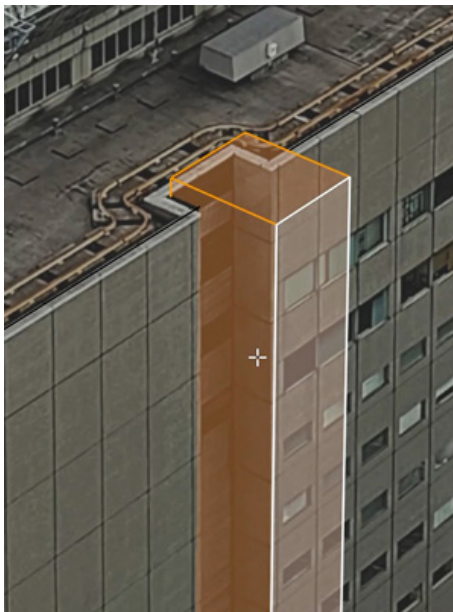


Now with the extruded Faces selected, go to **Face > Intersect Boolean**. This cuts a nice chunk out of our building.



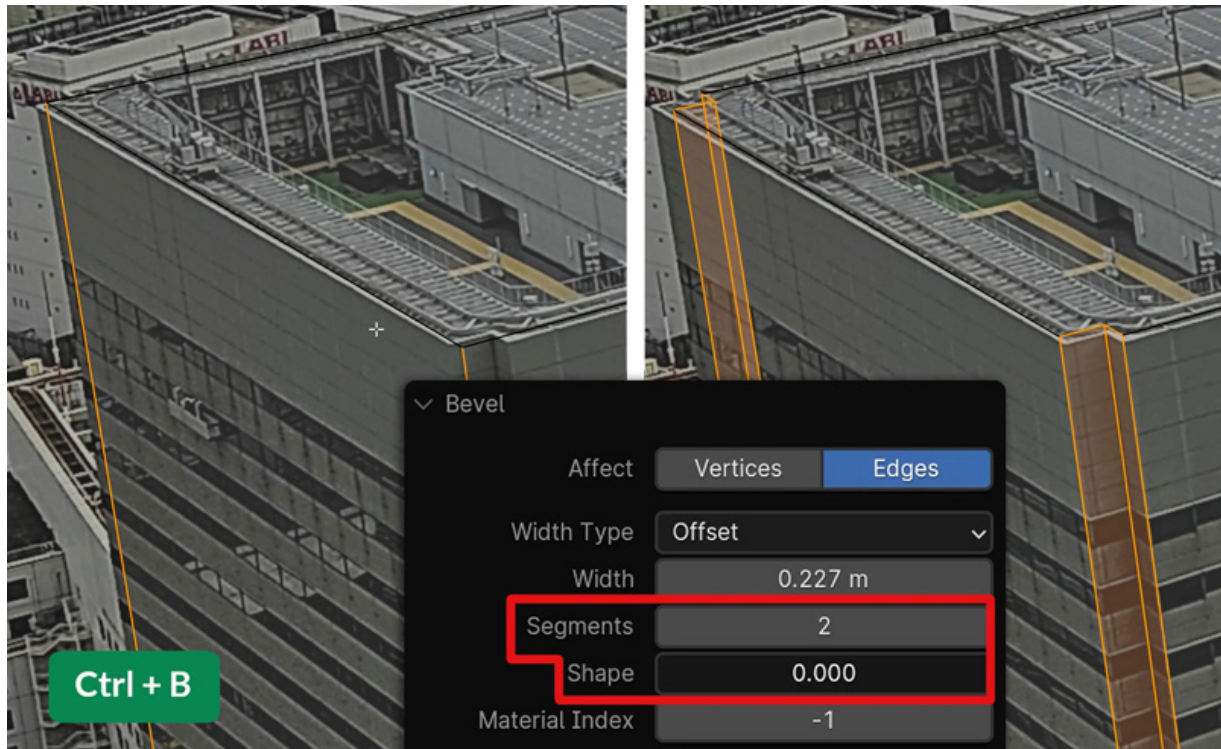
We can repeat the same steps on the other side. The benefit of duplicating a Face for this purpose, is that the Face is already perfectly aligned with the side we want to cut out of.

It seems the boolean cut didn't work on this side (see middle picture). If that happens, you can try a different **Solver** method or checking **Self Intersection**, which solved the problem in this case.

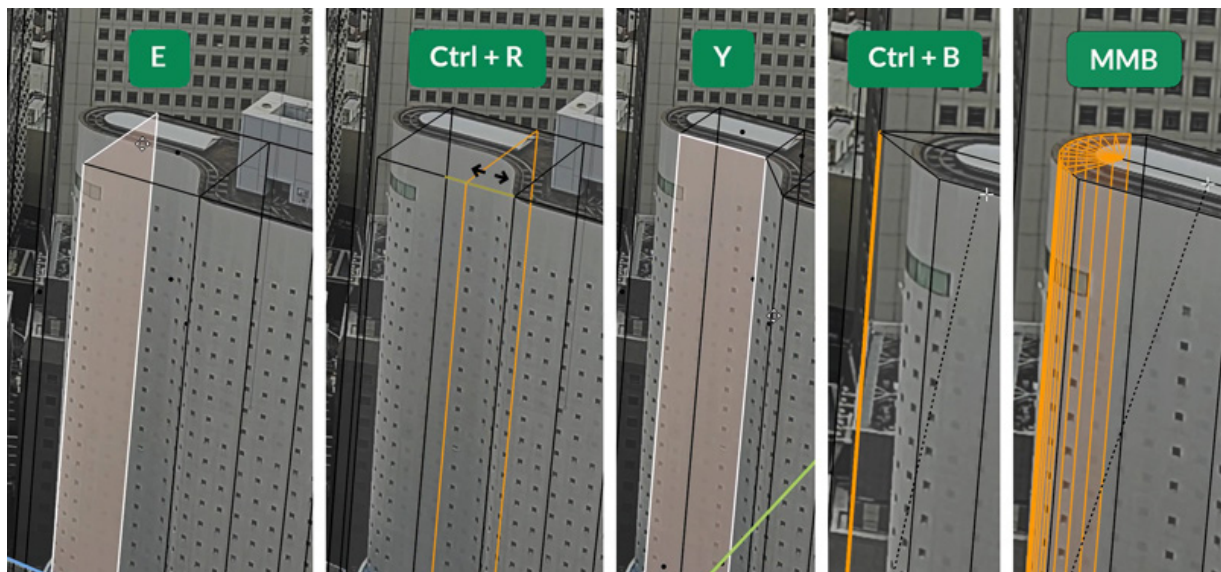


Some minor adjustment, by moving some vertices here and there, may be necessary. Next let's do the corners of this building. **Hold Shift** and select the Edges.

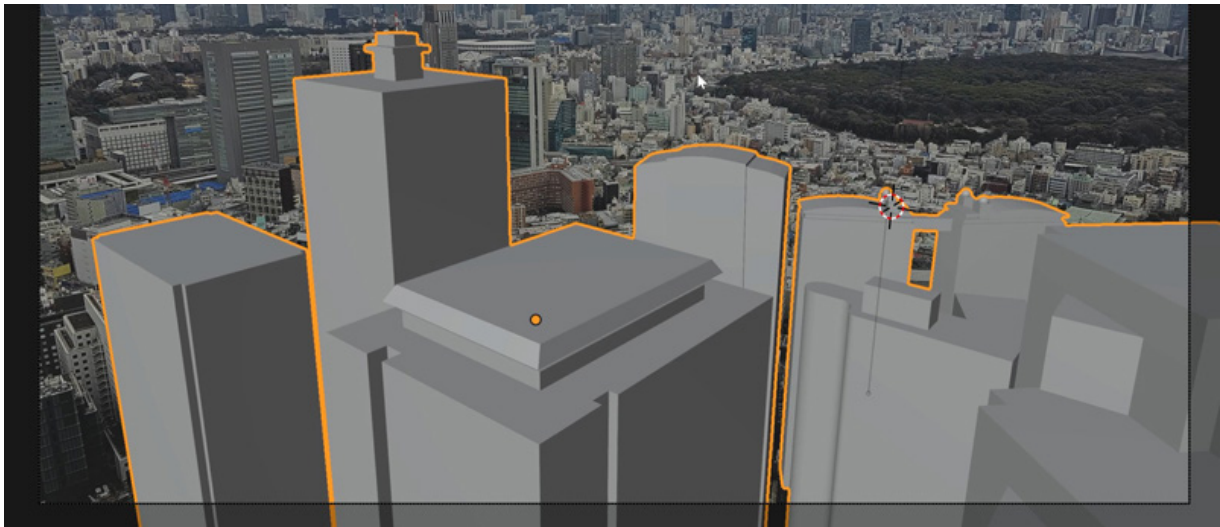
Press **Ctrl+B** to bevel, and **scroll the mouse wheel up** to add one segment. Changing the **Shape** to 0 gives us the exact shape we need.



We can fine-tune it if necessary. I won't explain the rest of the modeling step by step, as it's mostly very basic and repetitive stuff. Just lots of extruding, adding more Edge loops where needed, moving Faces, and beveling here and there to add some rounded parts.

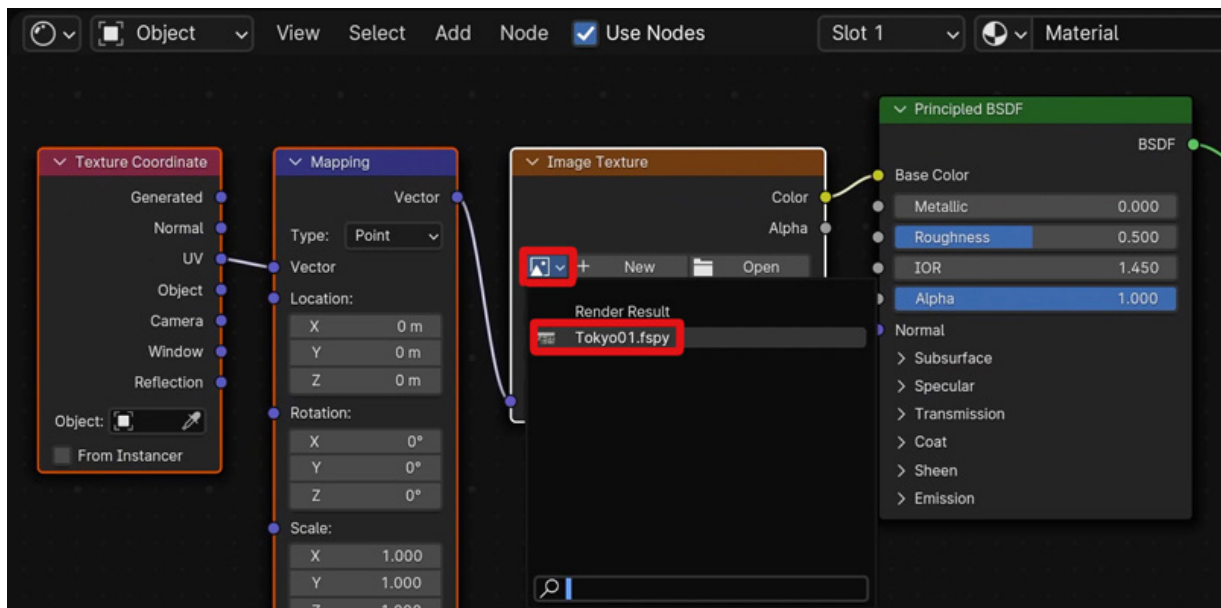


Honestly, a lot of these things are just simple cubes. You can get away with a lot when using projection mapping like this, because the texture has so much information. After some more basic modeling, we have a bunch of simple buildings.

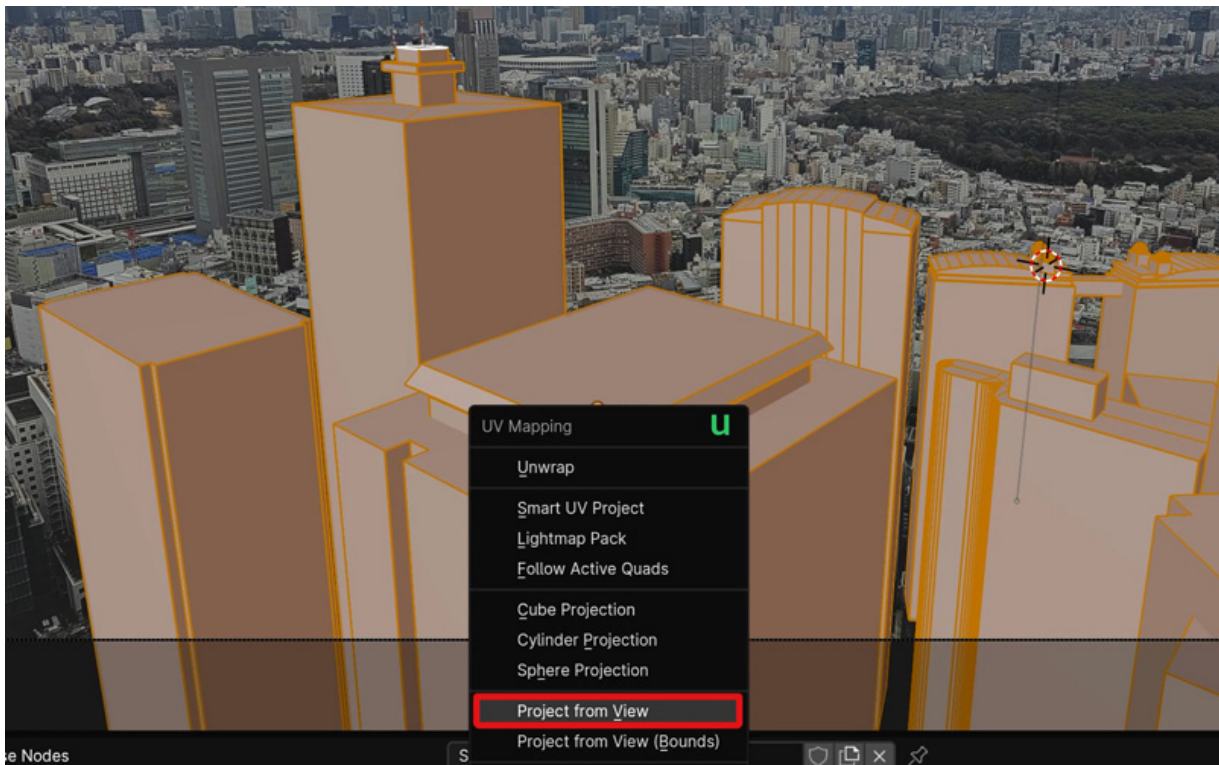


Next we need to add a material. I'll open a **Shader Editor** for that. Select the Principled BSDF and **press Ctrl+T** to add the necessary nodes (*Note that you'll need to enable the add-on Node Wrangler in Preferences to have this option*).

The photo is already loaded with the fSpy file, so we can choose it as the texture.

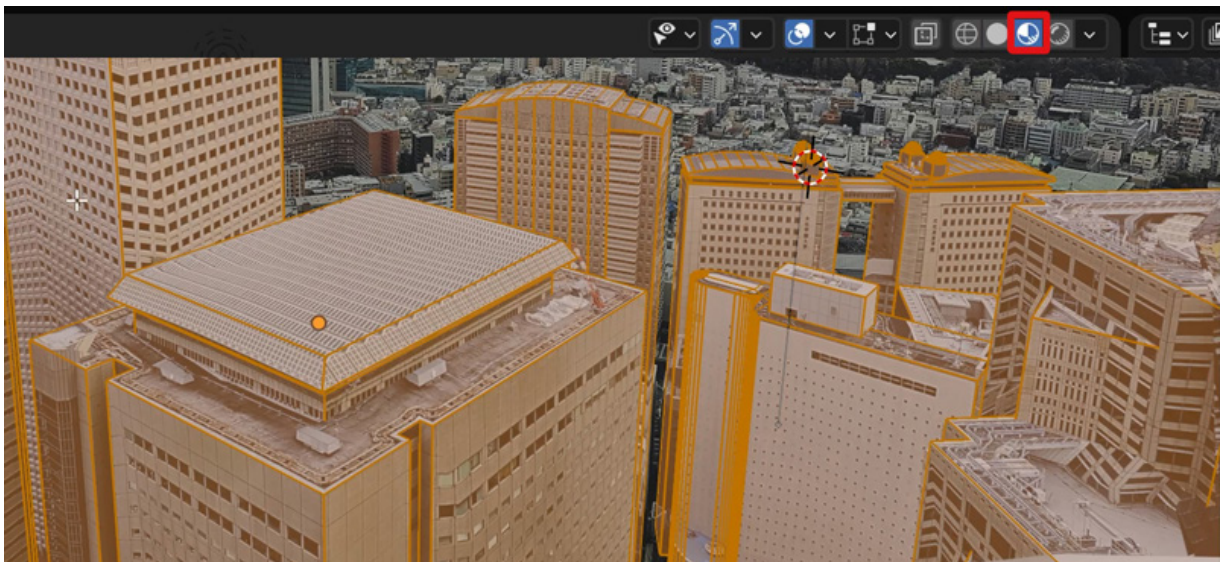


Select all in Edit Mode, **press u** and choose **Project from View**. This projects the texture from our current view, the Camera view.

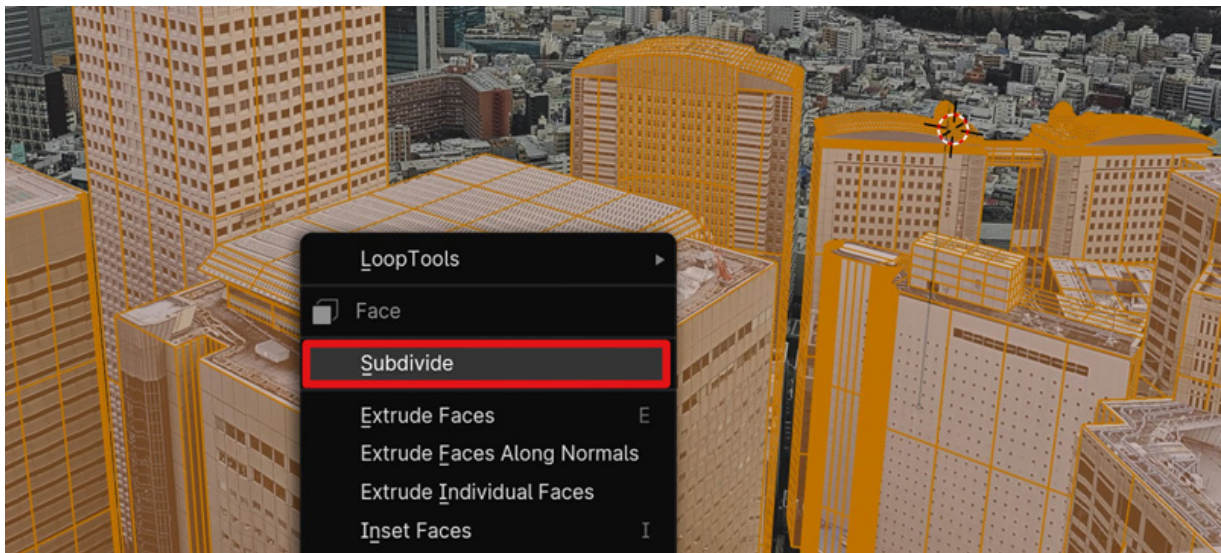


And switch to **Material preview** to see the result.

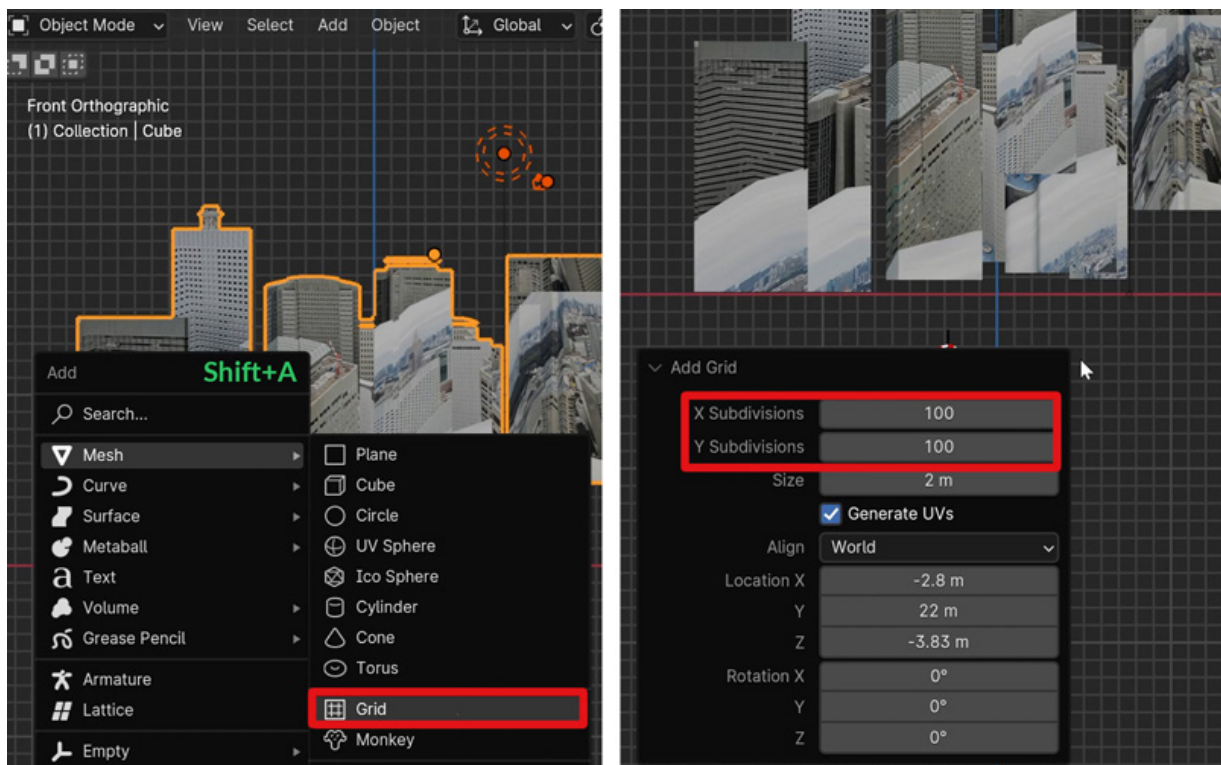
It kind of worked, but as you can see it's all wobbly and distorted. That's because there's not enough geometry to support the texture.



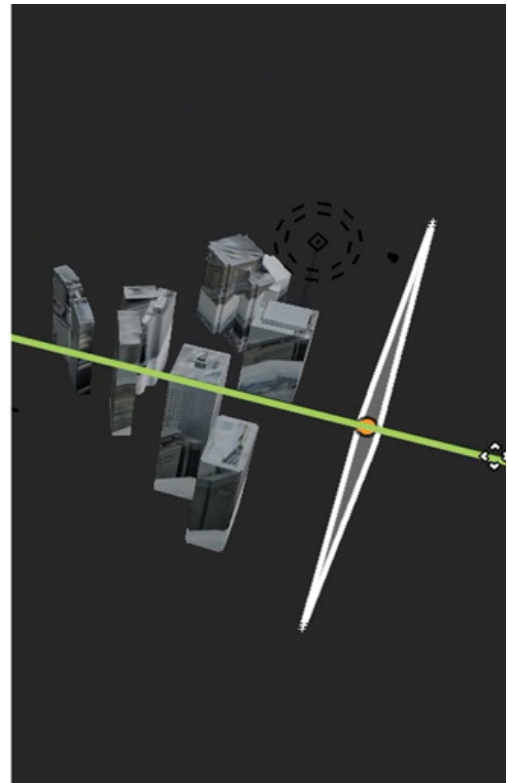
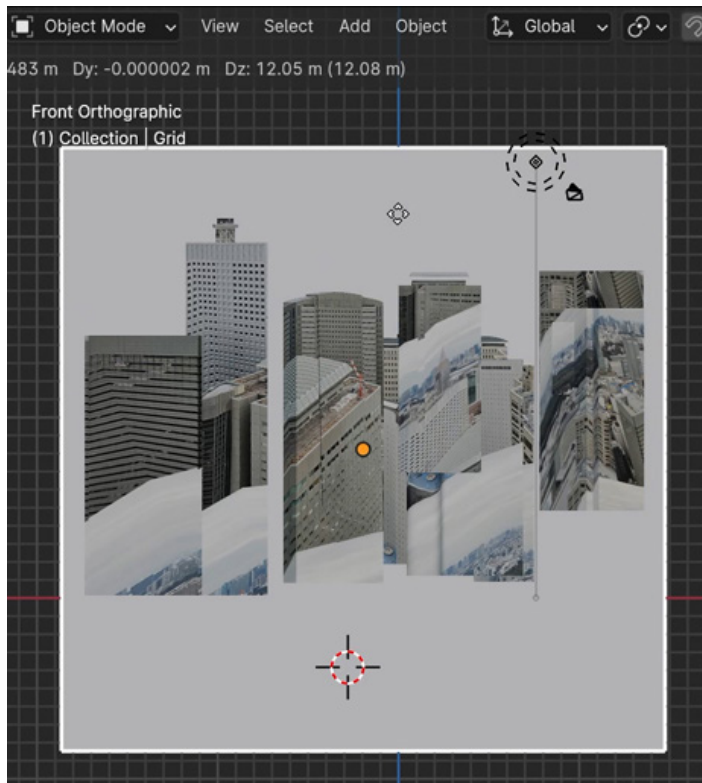
We can make this better by right-clicking and **Subdividing** a bunch of times, although in this case that adds a lot of unnecessary geometry in some places. So let's not do that (in some cases this can be a good solution, though).



Instead, let's "slice" the geometry. To do that, we'll add a **Grid**. And increase its resolution to **100 by 100 Edges**.

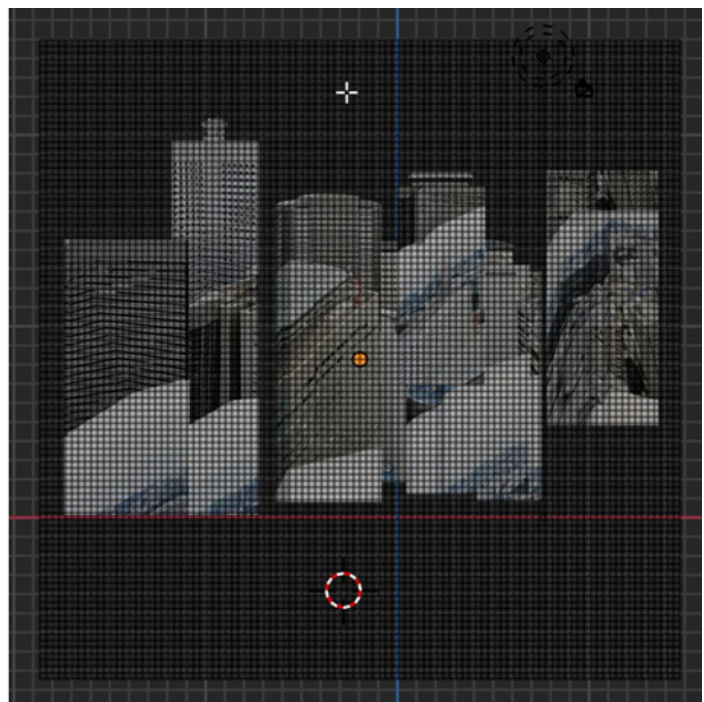
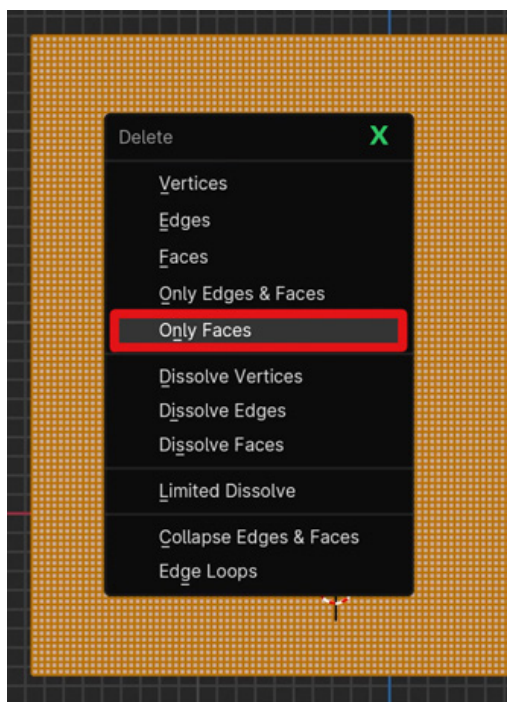


Rotate and scale it so it encompasses the buildings. And move it so it's not overlapping or inside the buildings.

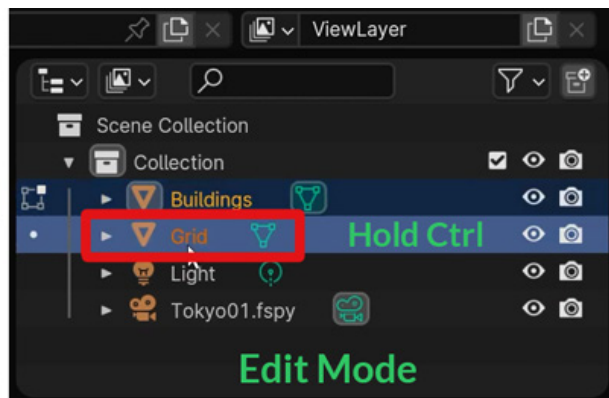
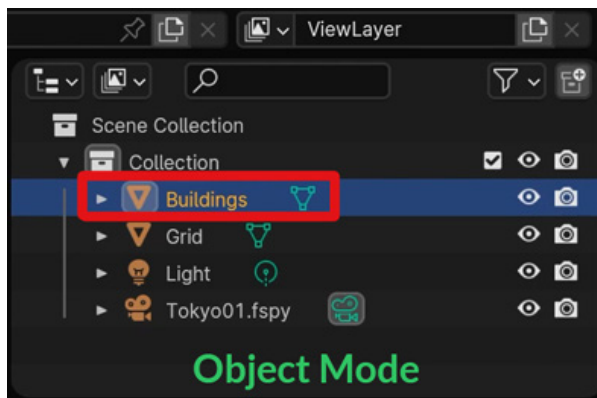


Go to **Front orthographic** view.

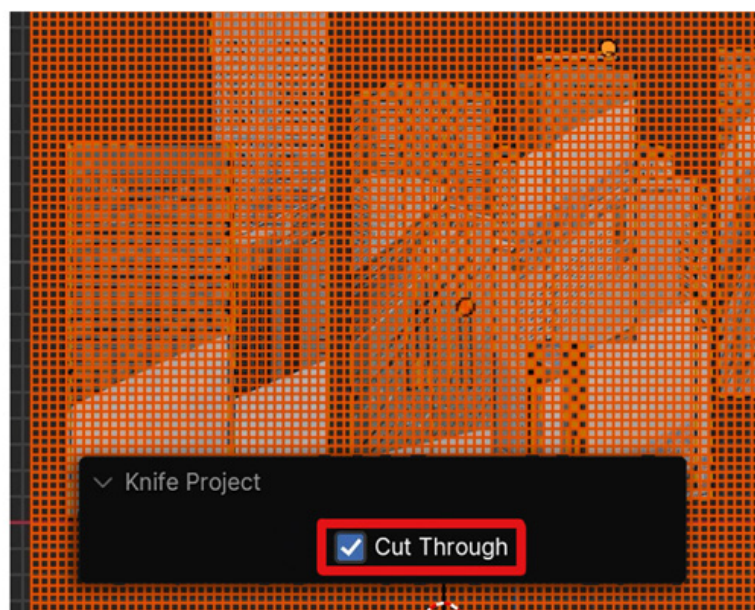
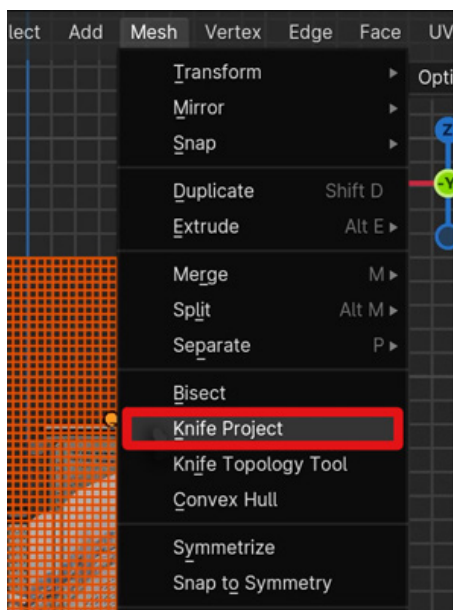
In Edit Mode press A to select all and press X, then choose **Only Faces**. Now we just have this grid of Edges.



Rename the buildings geometry to Buildings, to keep things organised. Now with the buildings selected in Object Mode, switch to Edit Mode. **Hold Ctrl** and in the outliner, select the Grid.



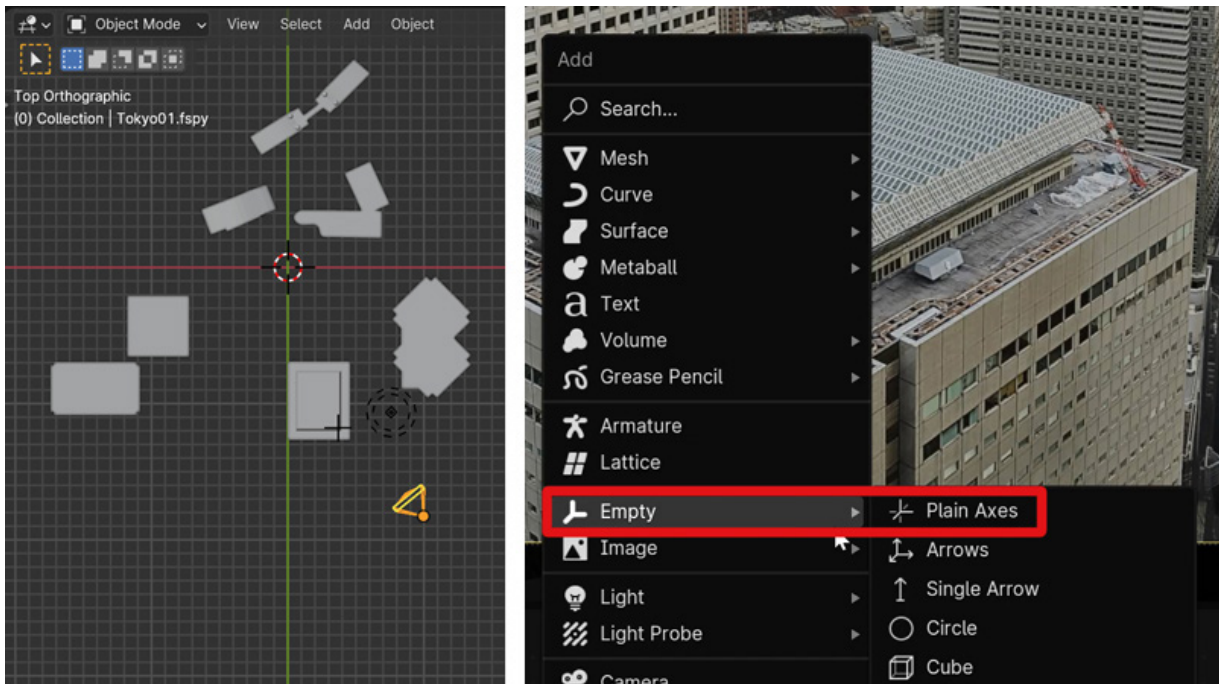
Go to **Mesh > Knife Project**. And then make sure **Cut Through** is enabled. You may have to wait a few seconds after enabling Cut Through, for it to work.



Now the buildings are 'sliced', giving them all enough geometry so that the texture is not distorted.

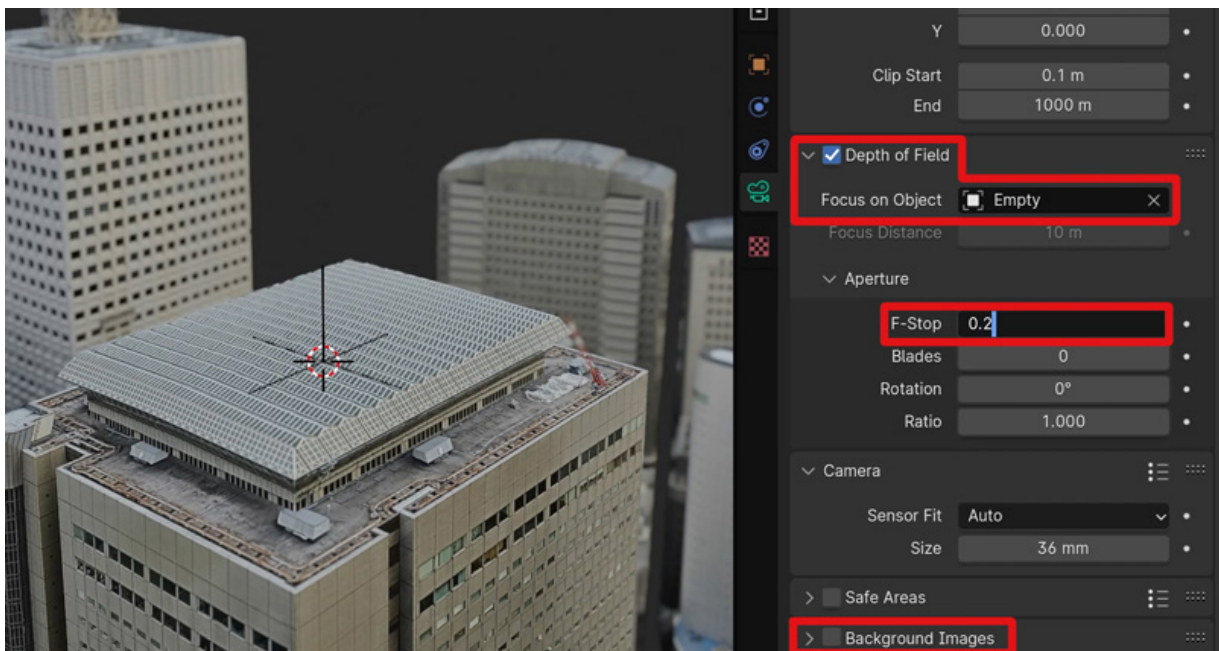


We can hide or delete the Grid now. Now if we carefully **move the Camera**, you see that we get a subtle camera move. And we can add an **Empty** to use as the **Depth of Field** target.

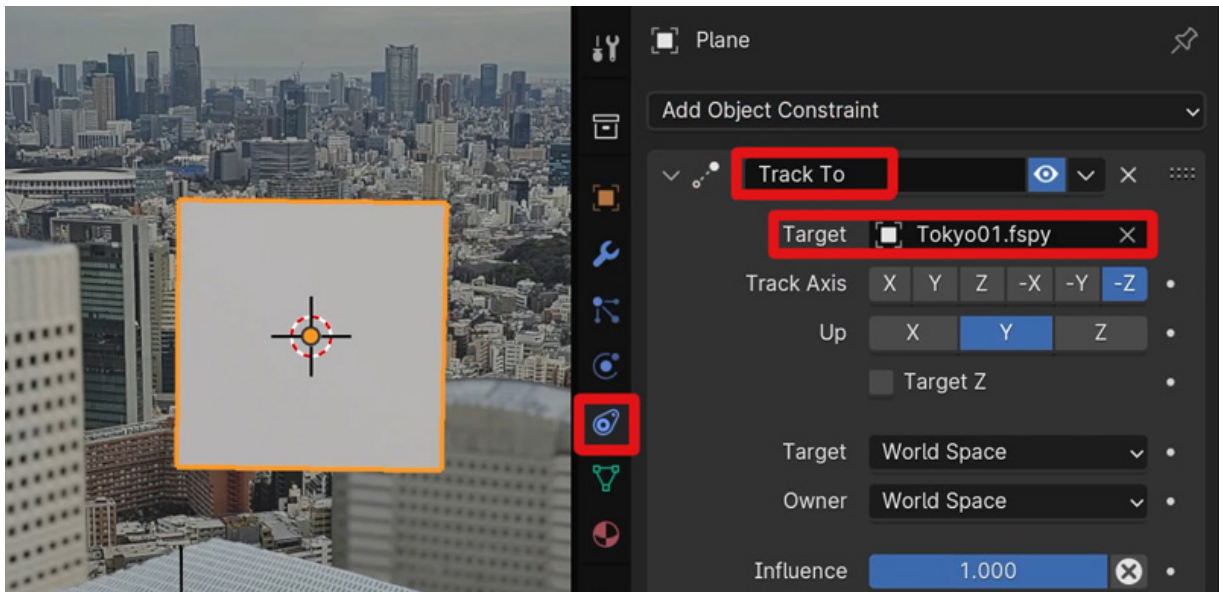


This gives us some unrealistic but nice looking **shallow depth of field**.

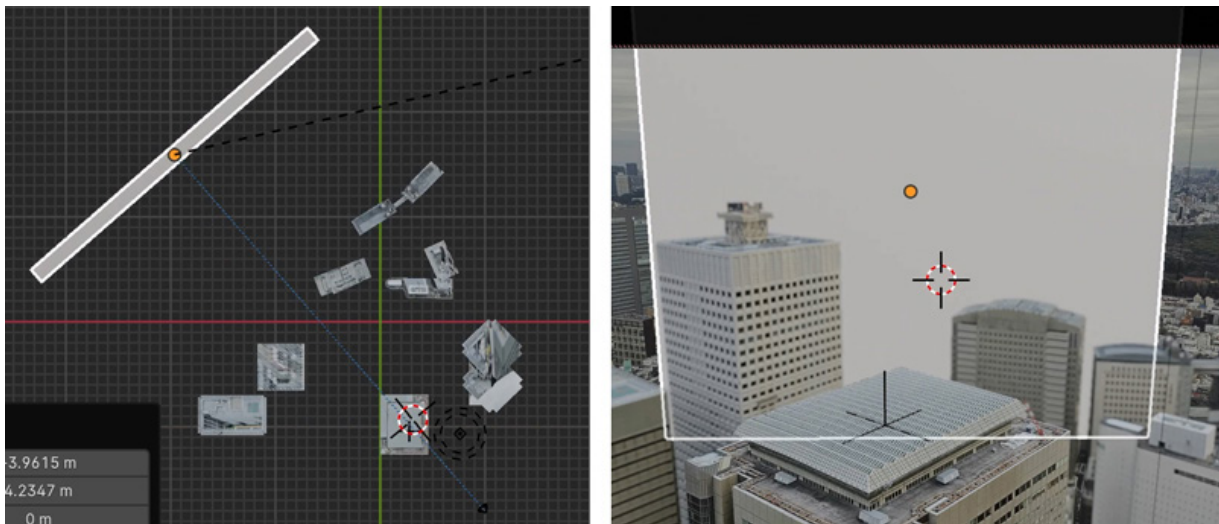
We can disable the **Background Image** now, as we no longer need it for reference. And it's not influenced by the depth of field anyway, which is not realistic.



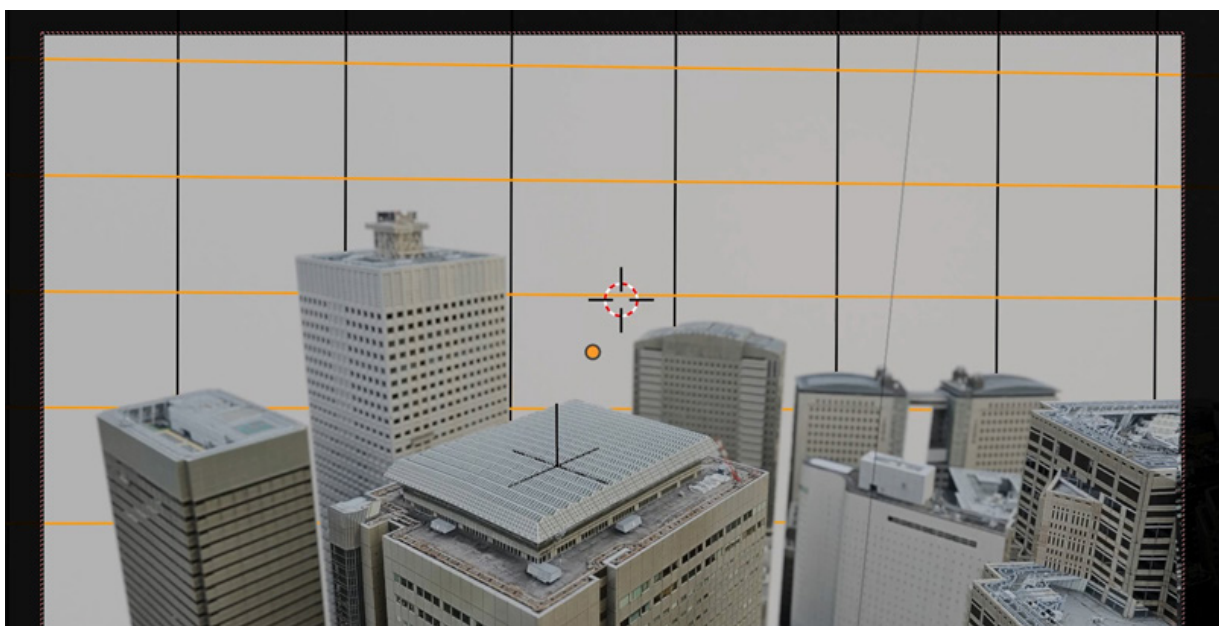
What we can do instead, is add a **Plane** and add the same texture to it. To make sure it's always facing the Camera, add a **Track To constraint** to it. Choose the Camera as its target.



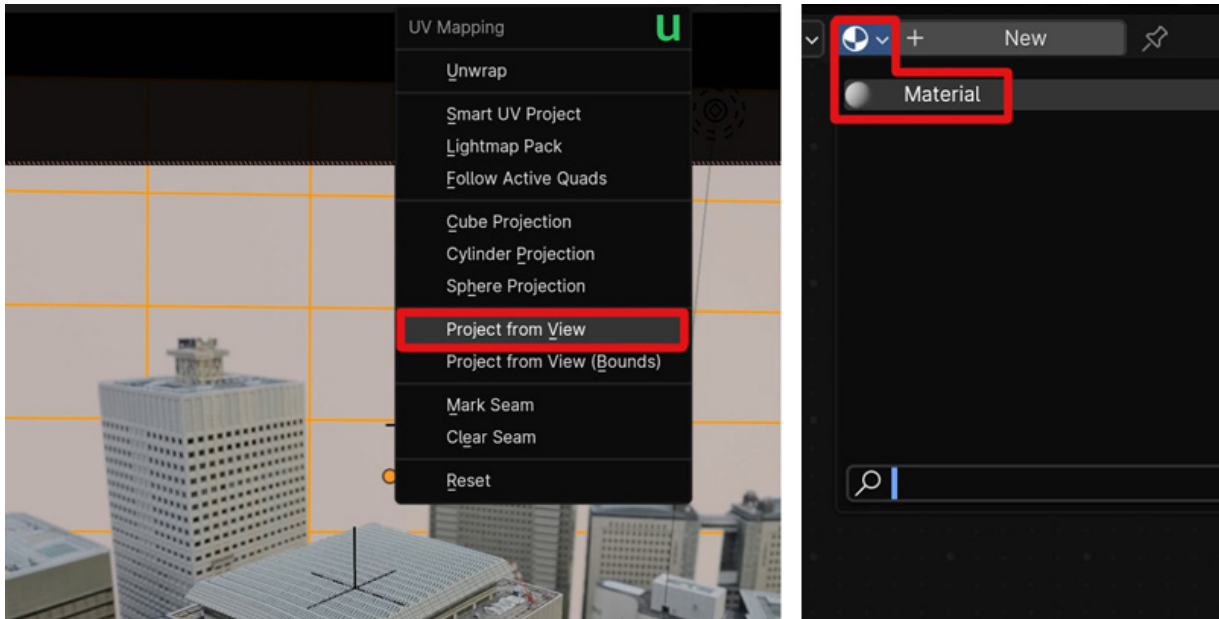
Then move the Plane behind the buildings and scale it up as necessary to fill the frame.



Add some Edge Loops (Ctrl+R) so that the texture won't be distorted.



And just like with the buildings, select all, **press U** and choose **Project from View**. Use the same material as the buildings.

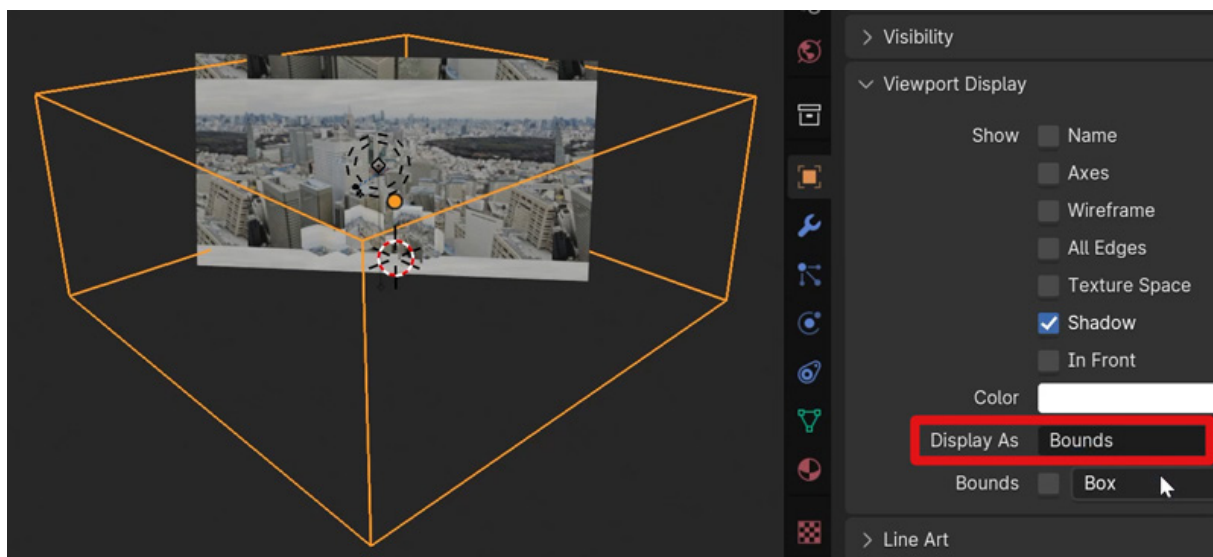


Now the far background is also affected by the camera depth of field. And we can move the Empty to change the focus point.



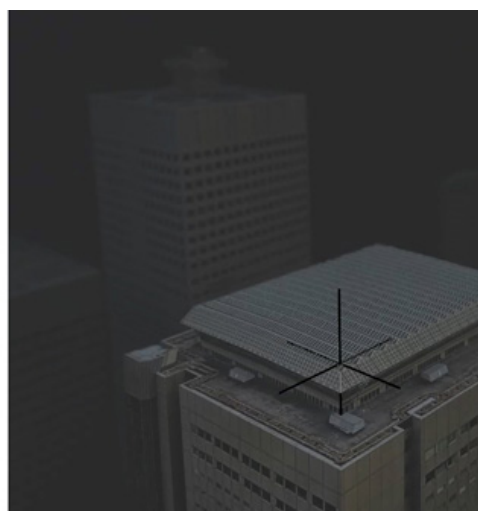
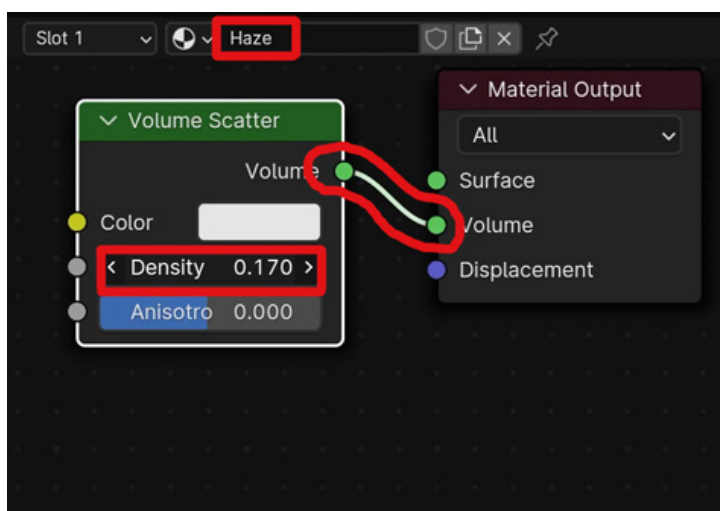
Now I'd like to add some **volumetric haze** to make it look more interesting.

To do that I just add a **Cube** around the scene. Scale it down to make sure it only envelops the scene, it doesn't have to be any bigger. And for ease of use I'll set it to **Bounds** in viewport display.



Give it a material with an appropriate name.

Remove the Principled BSDF and add a **Volume Scatter** node instead. Connect it to the Volume input of the material. The **Density** value controls how thick the haze is. The default value is too high, so you'll need to lower it to your liking.

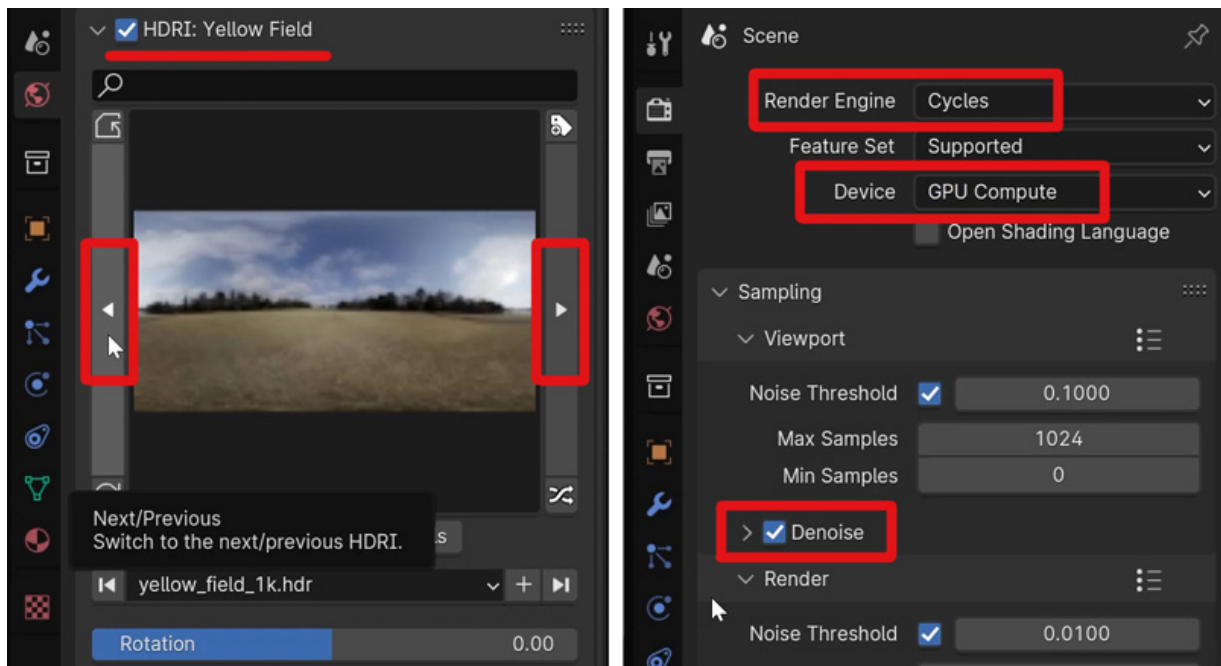


I'll add an HDRI using the [Gaffer add-on](#).

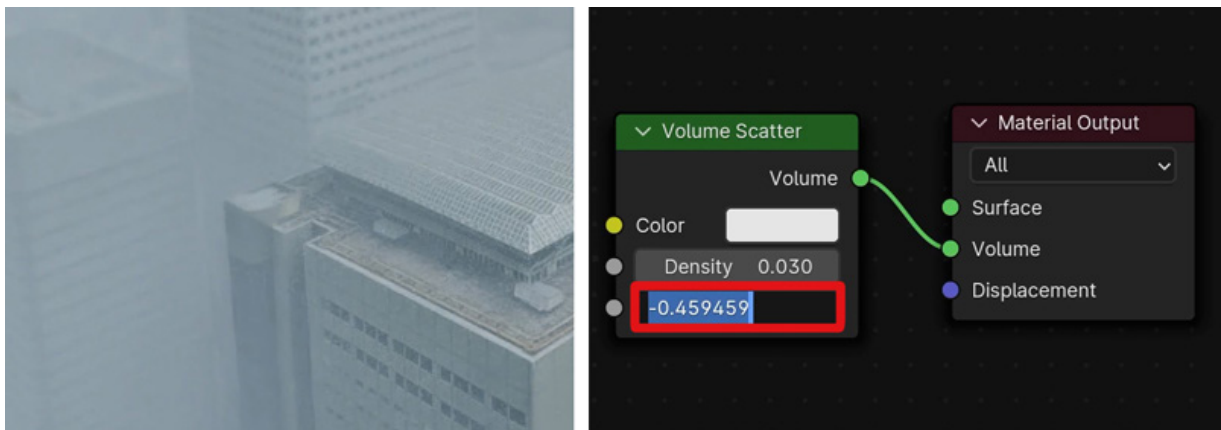
Of course all this will look much better in **Cycles**. And it will be faster if I use my **GPU** and enabled **Denoise**.

You don't have to use an add-on to add an **HDRI**, you can just do it [the old-fashioned way](#).

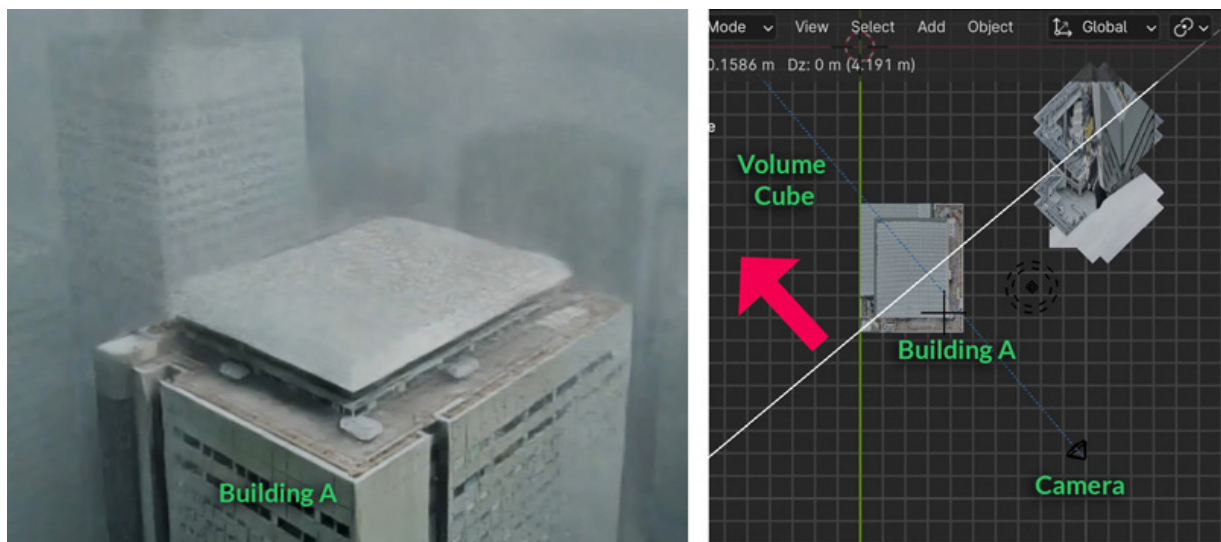
However, I really recommend [Gaffer](#) (made by Greg Zaal, of [Polyhaven](#) fame). It's a lot of fun to use. There's a [free](#) and [paid version](#) (so you can try it out - but if you like it, do consider supporting the developer). The many HDRI's of Polyhaven are also free.



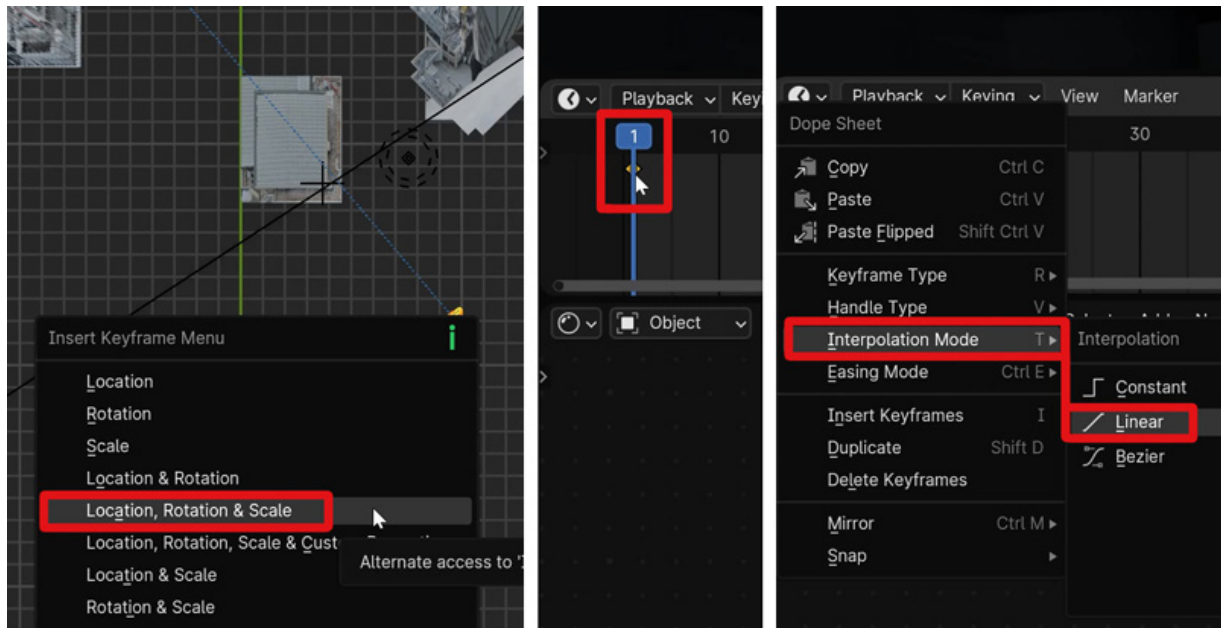
To control the look of the haze, we can use the **Anisotropy** value as well. I think it's a lot of fun to play with these kinds of values, as you quickly get different looks.



However you can also just move the **volume cube**. If you place it just before the Camera like this, the buildings nearest the Camera will have a bit more contrast.



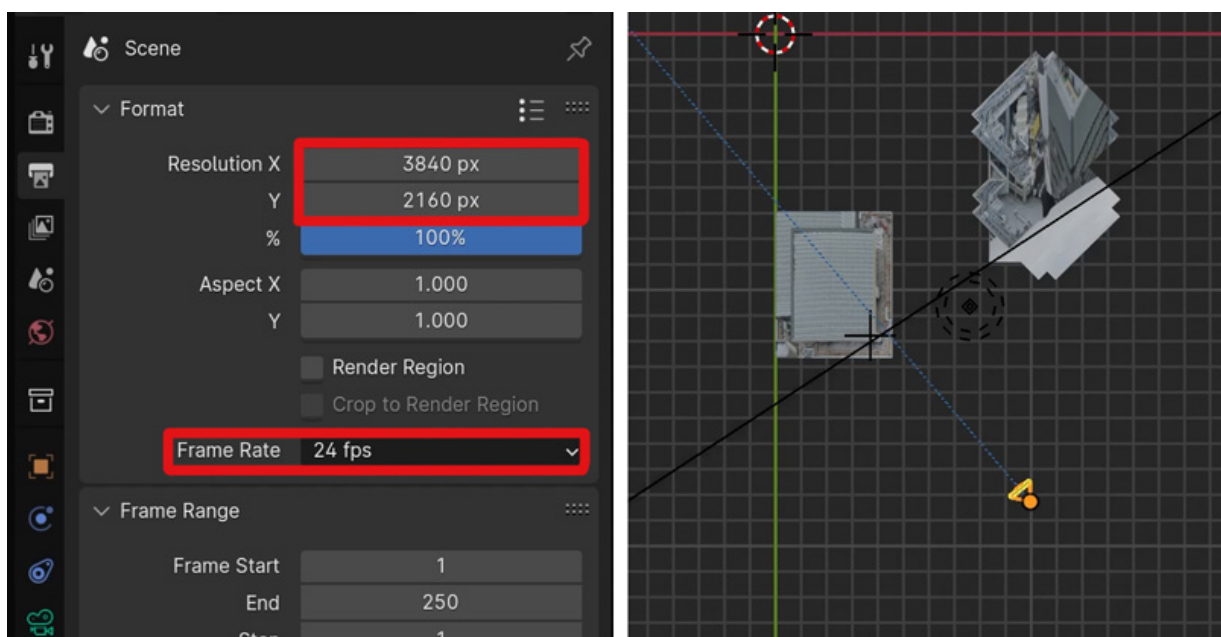
Let's open a **Timeline** so we can animate the Camera. Select the Camera and add a keyframe at frame 1 by **pressing i** and choosing **Location & Rotation** or **Location & Rotation & Scale**. Right-click on the keyframe and set the key to **Linear interpolation** so the movement is linear instead of easing in.



I'll change the resolution to a more standard 4K format. When using fSpy, it will set the scene to a strange resolution. 4K is **3840 x 2160 pixels** (1920 x 1080 multiplied by 2).

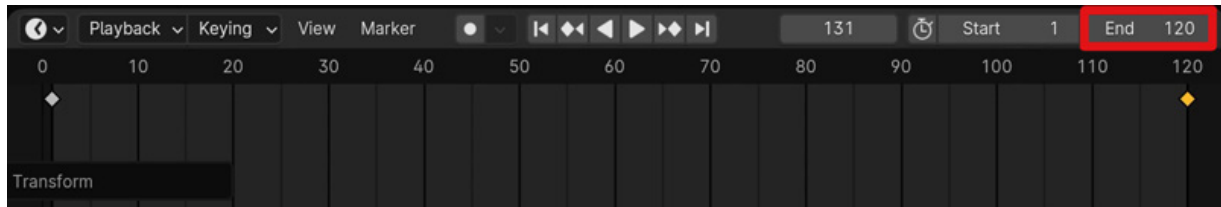
And then we can go to a later frame (frame 120 for example) and move the Camera to a new position by pressing G. By holding Shift, it's easier to make a subtle change in the location of the Camera. I recommend doing this in the **Top Orthographic view**.

You can't go too crazy with the camera position before you start seeing weird stretched parts of the texture, so it's best to keep the camera move subtle. **Set another keyframe.**

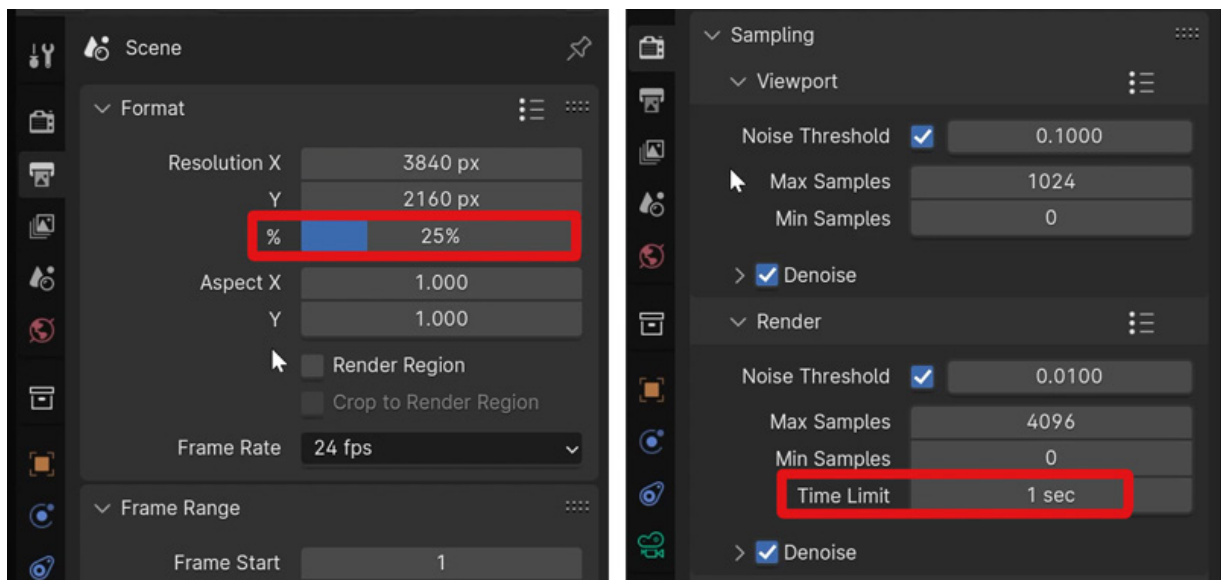


And I'll set that second key to Linear as well (by right-clicking on it and choosing Linear interpolation). If this was shot from a helicopter for example, the movement would be Linear in real life as well.

Let's calculate how many frames we need. I want **5 seconds** of animation with 24 frames per second, so that's **120 frames**. I'll make sure the second keyframe is at frame 120 and set the End to 120.

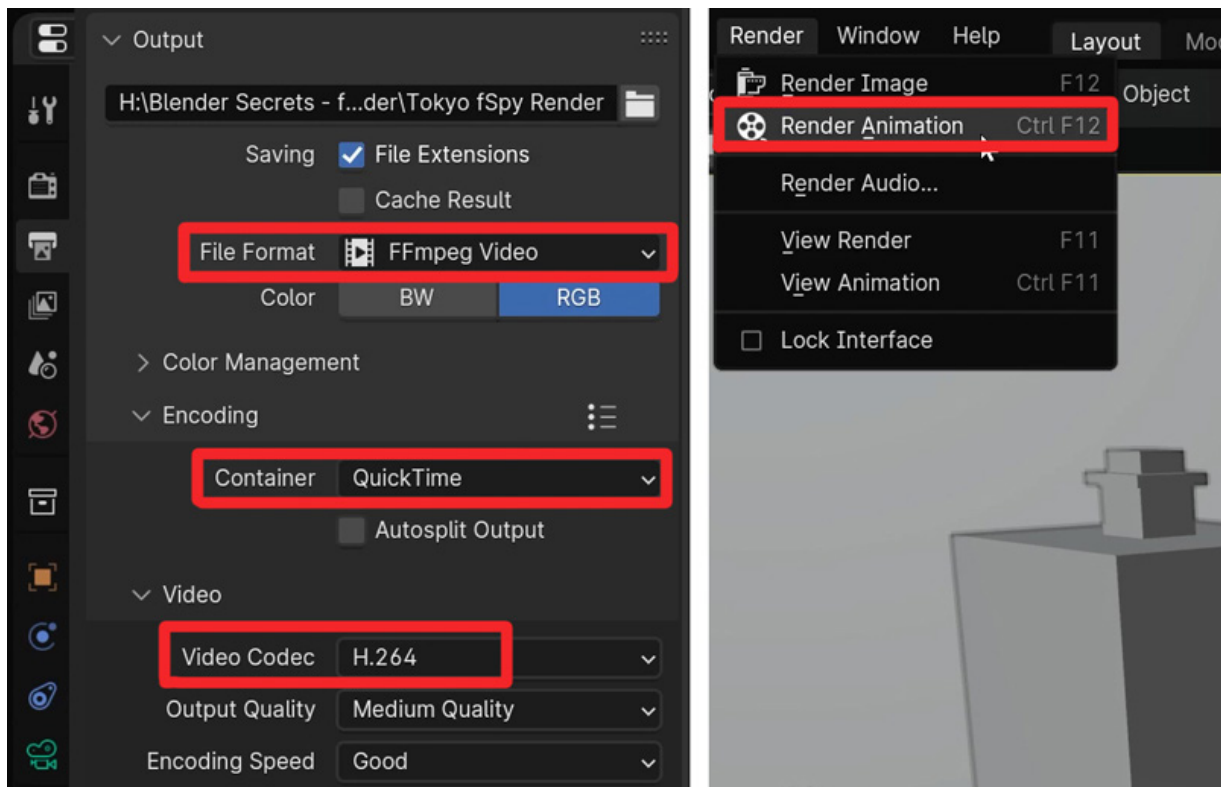


To do a test render, I'll limit the render time to 1 second per frame. And set the resolution to 25%.



As it will render fast, I will choose an **H264** video for the test.

Then **press Ctrl+F12** or choose **Render Animation**. At this low resolution and with only 1 second as the render limit, it goes pretty fast.

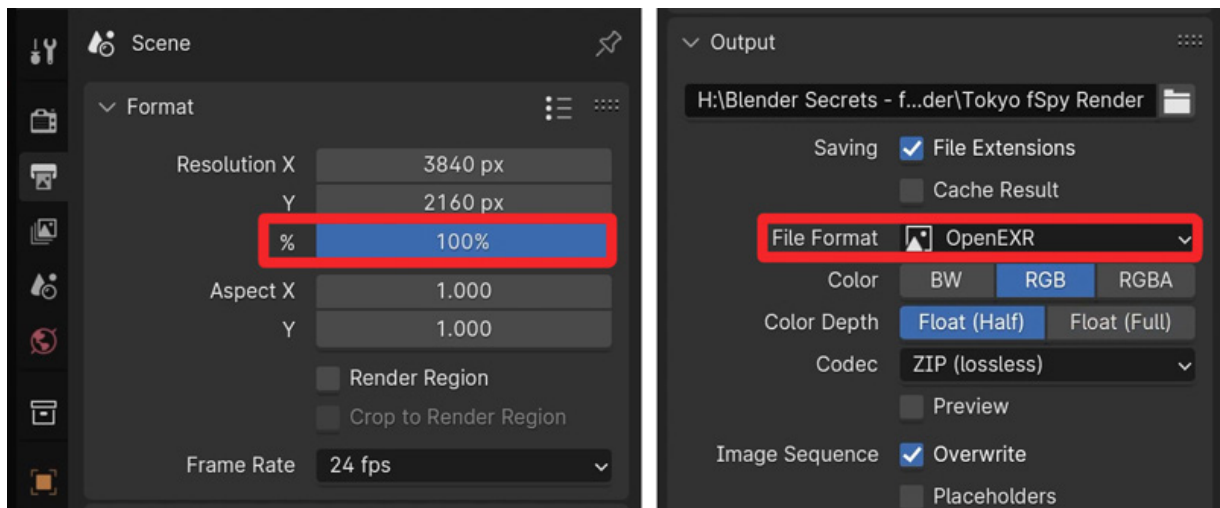


And after a few minutes, we have the result of the low quality test render that we can play back. I recommend always rendering a low quality quick render before you commit to rendering high quality frames.



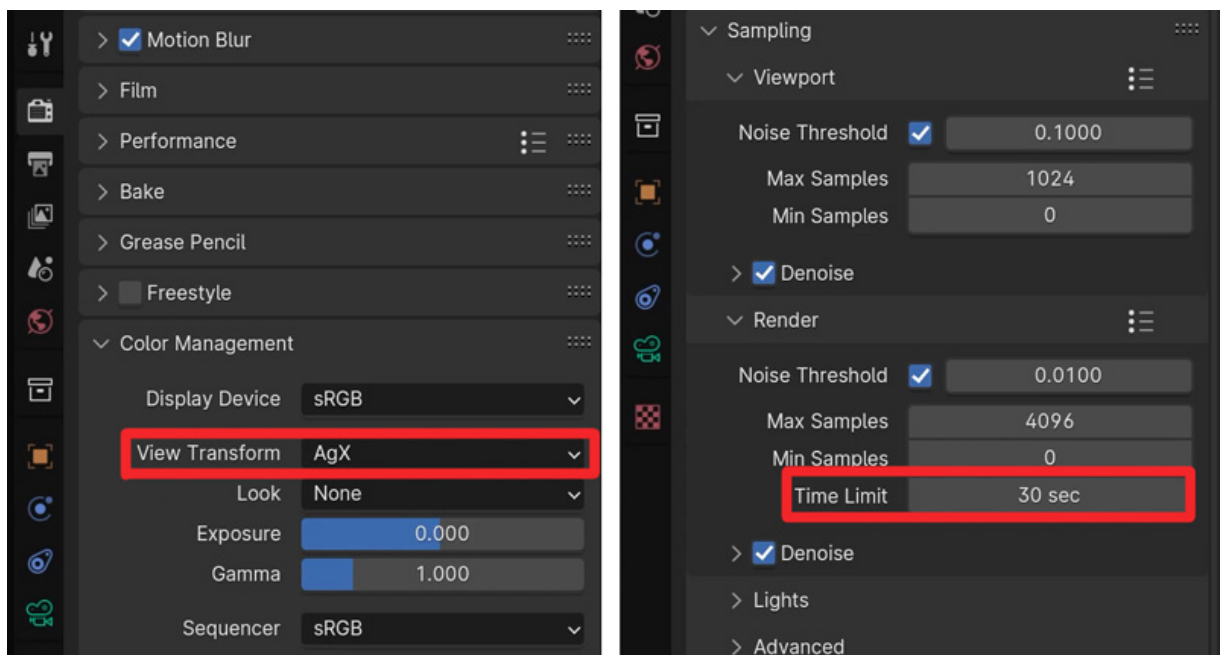
If the result is not to your liking, adjust the keyframes to make it more interesting. Once you've got something that you're happy with, you can set the resolution back to 4K for the final render. And **OpenEXR** for the file format. We'll render an **image sequence**. 16 bit, or half float, is enough to be able to do some nice color grading.

The advantage of rendering an image sequence instead of a video file, is that when the render stops at some point (for example, because of a computer or software crash), you won't lose the frames rendered up to that point. And you can just continue rendering from where it left off.



Just checking that it's set to **AgX**, which will give us a nice Linear image for later color grading. I'll also increase the **Time Limit** for the render. 120 frames times **30 seconds**, divided by 60 seconds gives us 60 minutes of render time.

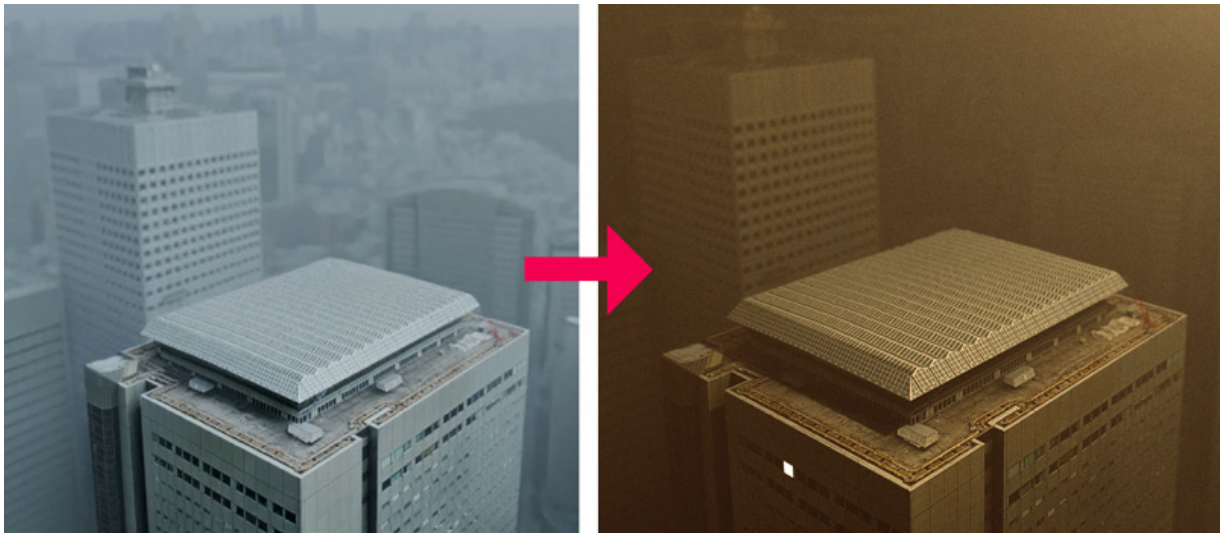
Before rendering, it's good to **save** just in case. In reality it took about 3 hours and 20 minutes to render all the frames, so the Time Limit isn't very accurate.



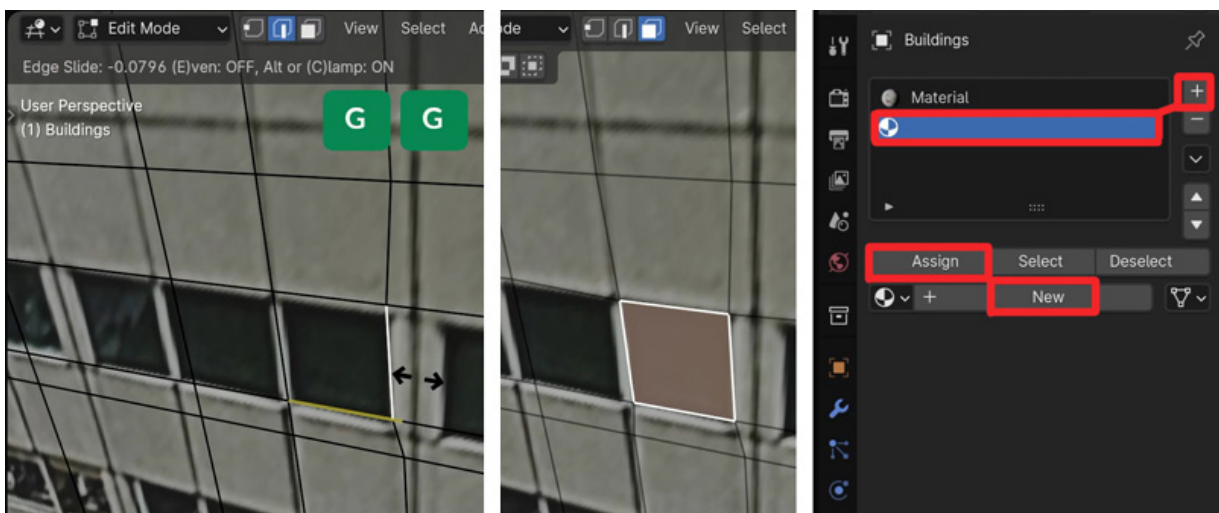
And to make it easier for the computer to render all this, I'll set the viewport to Solid mode.

But actually, before I render I want to make the scene a bit more mysterious.

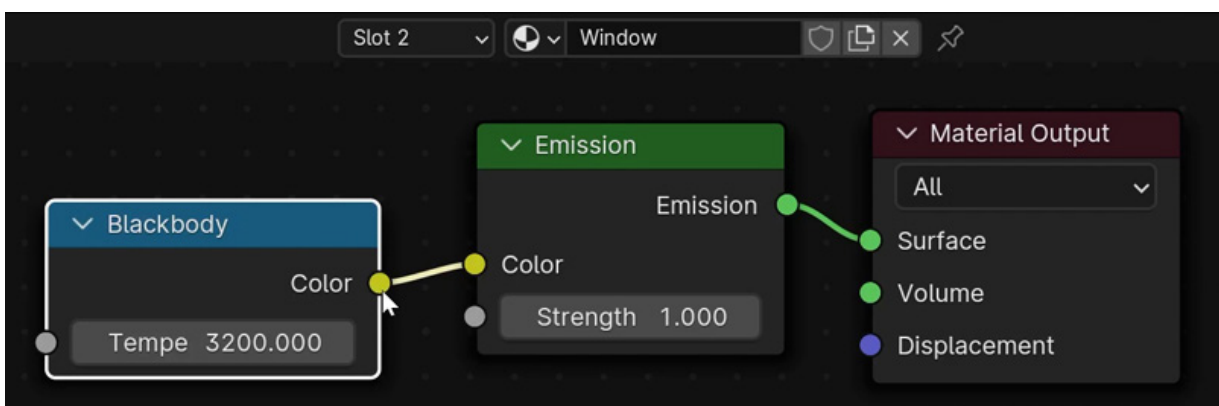
I think it could be interesting if one of these windows had light inside, as if that's the last person in the whole city living there.



I'll need to add a couple more Edge Loops so I can select just that window. **Press Ctrl+R** to add Edge Loops where needed. If you **press G twice** you can slide Edges without distorting the texture. I'll add a second material to the window. Assign it to the selected Face and give it a name.

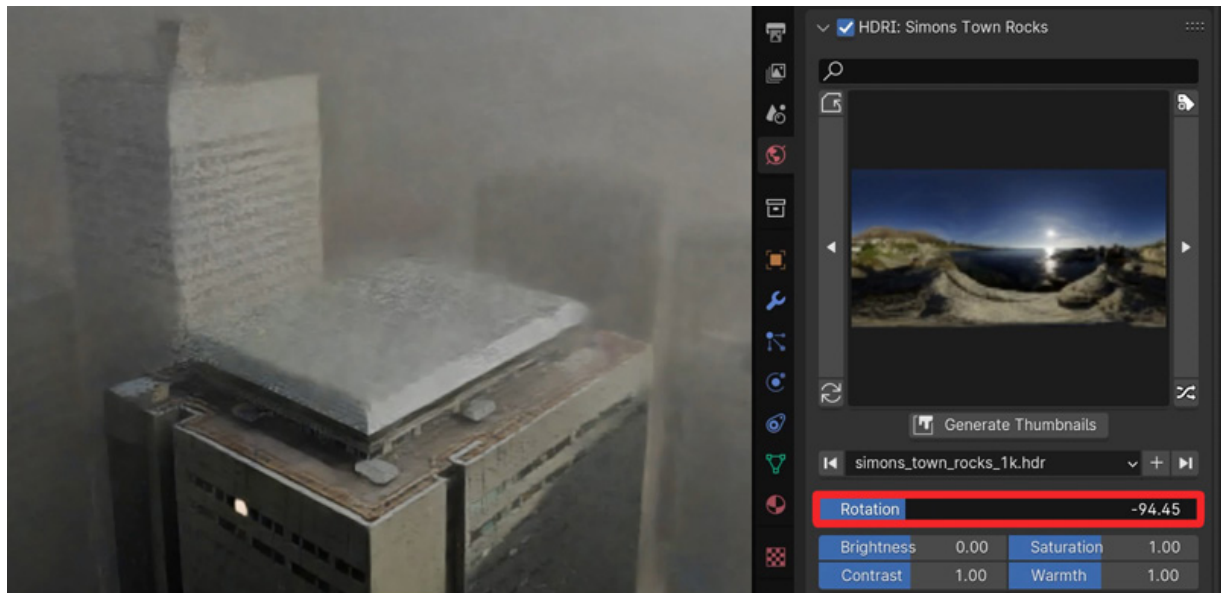


Remove the Principled BSDF and put an **Emission** shader in its place. We can manually give it a color... but to be more realistic, we can add a **Blackbody** node to use a real-life color temperature value. Something like 3200 Kelvin gives a nice warm artificial light color.

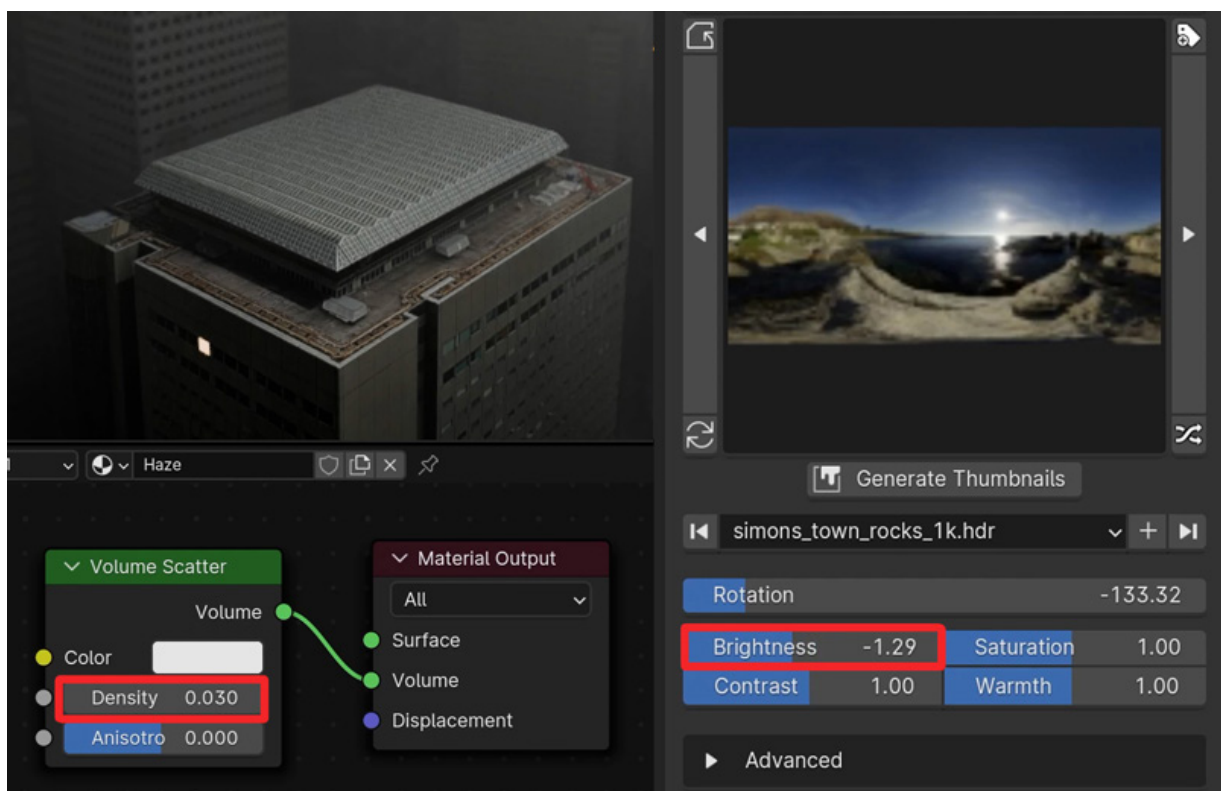


You can google these more common blackbody values. Looks like **3400 Kelvin** would be even more appropriate.

I'd like the light to stand out more. We can try to increase the **Strength**... or we can experiment with the lighting, so the window is in the shaded side of the building, which makes it stand out more. We can try different HDRI's, or by **rotating the HDRI** we can also get very different results.



We can make it even darker and more dramatic by increasing the **Density** of the Volume. We can also reduce the **Brightness** of the HDRI. This looks quite interesting and mysterious. Let's save the .blend-file and then render the animation in 4K.



And here is the result of the render, after some quick color grading.



If you want, you can also download the video [here](#), so you can watch it offline without ads.

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